

The Impact of Offshoring and Import Competition on Carbon Emissions*

Jonathan Leisner Jakob R. Munch August Twile Nielsen Georg Schaur

March 2026

Abstract

We use Danish firm-level data to jointly examine the causal effects of offshoring and Chinese import competition on carbon emissions, output, and the emission intensity of output. Offshoring reduces firms' emission intensity but also increases their production such that firm-level emissions do not change. Chinese Import competition reduces firms' production and their emission intensity. Next, we apply our firm-level estimates in counterfactual decompositions and in the computation of emission-leakage rates. The resulting evidence shows that offshoring leads to cleaner production in the manufacturing sector while import competition from China mostly leads to more emissions abroad.

(*JEL* F14, F18, Q54, Q56)

*Contact information: Leisner: University of Copenhagen, jl@econ.ku.dk; Munch: University of Copenhagen, jakob.roland.munch@econ.ku.dk; Nielsen: University of Copenhagen, aetn@econ.ku.dk; Schaur: University of Tennessee, gschaur@utk.edu. We thank Morten Bennedsen, Jevan Cherniwchan, Rikard Forslid, Alessia Lo Turco, Luca Macedoni, Joel Rodrigue, Bertel Schjerning, Joe Shapiro, Reed Walker and seminar and conference participants at SNEE Malmø, OASIS, Midwest International Trade Conference Purdue, Berkeley, DIEW Aarhus, SETC Sardinia, LSE Environment Week, EWMES Berlin, Bank of Canada and LMU Munich for helpful comments. Thanks also to Mathilde Vallat for excellent research assistance.

1 Introduction

As carbon emissions associated with manufacturing fall in rich countries, they increase in developing countries (Copeland, Shapiro, and Taylor 2022). At the same time, two dominant trends in globalization, import competition in final goods and the sourcing of intermediate inputs from low wage economies, have a potentially substantial impact on the location of production (Autor, Dorn, and Hanson 2013; Johnson and Noguera 2012). However, causal evidence on the effect of international trade on local and global environmental outcomes is surprisingly limited (Najjar and Cherniwchan 2021). This paper disentangles the potential joint effects of import competition and offshoring to examine if and how they contribute to the local and global environment.

We do so in four steps. First, we start by combining Danish firm-level register data for the manufacturing sector, Industrial Energy Use Survey information, and firm-level trade data into an estimation sample that includes firm-by-year observations of output, energy use by source, and input based carbon (CO_2) emissions in addition to constructed measures of offshoring and import competition (Hummels et al. 2014; Bernard et al. 2025). Second, we develop shift-share instrumental variables that capture variation in comparative advantage (Autor, Dorn, and Hanson 2013; Bartik 1991; Borusyak, Hull, and Jaravel 2021) to identify the joint effects of offshoring and import competition on firms' output, emission intensity measured as emissions per unit of output, emissions, and we provide evidence for several underlying mechanisms. Third, using our instrumental variable estimates, we provide decomposition and counterfactual-decomposition evidence to evaluate the effect of offshoring and import competition on the aggregate emission intensity in the manufacturing sector.¹ Fourth, again using our elasticity estimates, we provide evidence regarding the consequences of import competition and offshoring for local and global emissions.

Over our sample period from 1996 to 2016, our data and the existing literature document

¹Cherniwchan, Copeland, and Taylor (2017) note that while decompositions of emission levels have been influential, they provide little information regarding micro-level adjustments. Instead, they proceed to decompose aggregate emission intensity into firm-level emission intensities and firm-level output. Following their approach, we focus on the manufacturing sector's emission intensity to examine firm-level adjustments.

a substantial surge in Danish imports from China (Autor, Dorn, and Hanson 2013).² Our empirical approach provides evidence that import competition from China in final product markets and offshoring with the rest of the world have clear but distinct effects for the environmental performance of Danish firms. Offshoring reduces firms' emission intensity but raises output such that emissions are not significantly affected. These effects are associated with firms' using more but cleaner energy, as well as improvements in energy efficiency and productivity in response to offshoring. Import competition from China also leads to lower emission intensities. In addition, it depresses output and emissions. These effects go hand-in-hand with firms' use of cleaner energy, improved energy efficiency, and lower energy consumption. That is, based on firm-level evidence, both import competition from China as well as offshoring potentially affect the aggregate emission intensity in the manufacturing sector via changing emission intensities within firms and the reallocation of sales across firms.³

Our decomposition and leakage evidence quantifies the relative importance of offshoring and import competition for the local and global economy. According to our counterfactual-decompositions, offshoring has substantially contributed to cleaner production in the manufacturing sector due to changes in firm's emission intensities as well as the reallocation of production across firms. Import competition has had almost no discernible effect on the aggregate emission intensity. Furthermore, our evidence on carbon leakage shows that import competition from China is associated with substantial carbon leakage compared to offshoring. Therefore, offshoring has a much more substantial effect on cleaning up local production while import competition with China mostly leads to substantial emissions abroad.

Our empirical approach and results provide several contributions to the literature. Aghion et al. (2024) emphasize the importance of jointly accounting for imports of firm-level interme-

²For example, in the data section we provide evidence that the increase in imports from China relative to domestic expenditure on goods is comparable to the U.S. where it has led to substantial effects in output and labor markets. For this reason, we focus on import competition from China and refer to it interchangeably as "Chinese import competition", "Chinese import penetration" or just "import competition" unless otherwise stated.

³Our data and existing literature also provide evidence for a surge in imports from the Eastern European countries included in the EU enlargement (Dauth, Findeisen, and Suedekum 2014). Despite the increase in manufacturing imports from these countries, we do not find evidence that this affected the environmental performance of Danish firms.

diate inputs and import competition in final product markets to avoid omitted variable bias. Addressing endogeneity in offshoring and import competition, we jointly identify the effects of offshoring and import competition from China, two major drivers of recent globalization experiences, and disentangle their effects on firms' environmental performance.

Copeland, Shapiro, and Taylor (2022) explain that most of the variation in aggregate emissions is due to within industry changes instead of changes in countries' industrial composition. This may be interpreted as evidence that standard comparative advantage mechanisms that explain industry composition play little role in explaining countries' environmental outcomes. However, our results imply that comparative advantage driven import competition measured at the firm level affects emission intensities and emissions across firms within the manufacturing sector.⁴

Furthermore, we also provide evidence that comparative advantage affects environmental performance via firms' offshoring activities. Although offshoring does not affect firm-level emissions, we provide evidence that offshoring due to changes in comparative advantage has relevant effects on the decline in the aggregate manufacturing emission intensity. In particular, we generate the hypothetical decline in the manufacturing emission intensity in the absence of import competition and offshoring and compare it to the observed decline in the manufacturing emission intensity. We find that offshoring has contributed to the overall decline in the manufacturing emission intensity through its effect on firm-level emission intensities and the within firm component. Furthermore, firms that grow due to offshoring tend to be relatively clean such that dirty firms are left with lower market shares, which in total leads to a lower overall manufacturing emission intensity. By contrast, although Chinese import competition causes exposed domestic firms to pollute less and therefore potentially affects total emissions, import competition does not contribute to improvements in the overall manufacturing emission intensity. This is because the least pollution intensive Danish manufacturing industries have experienced the largest inflows of final goods from China, including textiles and electronics, which tends to reallocate economic activity towards dirtier production. In other words, exposed clean firms shrink, while dirtier firms gain

⁴We note that Denmark is relatively well endowed with high-skilled labor. Accordingly, Denmark has a comparative advantage in products that are skill intensive. For example, there is a cluster of pharmaceutical companies including Novo Nordisk.

market shares. Thus offshoring and Chinese import competition, both driven by comparative advantage, have very different implications for the domestic manufacturing emission intensity.

Our final contribution is to assess the implications of offshoring and Chinese import competition for global emissions. Climate change is a global challenge caused by the global carbon emission level, and so it is important to assess the extent to which reductions in local emissions are offset by increased emissions elsewhere. We quantify the impacts of offshoring and Chinese import competition by calculating carbon leakage rates that incorporate our causal elasticity estimates of the effects of import competition and offshoring. To calculate the increase in emissions abroad associated with changes in final good imports and offshoring in Danish firms, we use product-by-country CO₂ intensities from around fifty countries. Not surprisingly, based on the magnitudes of the estimated leakage rates, Denmark’s production is relatively clean. A ton of emissions reduced in Denmark due to import competition or offshoring leads to a greater amount of emissions abroad.⁵ More importantly, the leakage rates provide evidence that import competition is associated with much greater carbon leakage than compared to offshoring activities.⁶ This is evidence that policies to address carbon leakage such as the Carbon Border Adjustment Mechanism should emphasize the harnessing of emissions associated with trade in final products.

As highlighted by Copeland, Shapiro, and Taylor (2022) there has been relatively little work that attempts to identify the causal effect of increased trade due to comparative advantage in affecting environmental outcomes, and only very few papers consider a global pollutant such as CO₂ as the outcome. Regarding the effects of offshoring at the level of firms, Dussaux, Vona, and Dechezleprêtre (2023) and Akerman, Forslid, and Prane (2024) estimate the effect of firm-level offshoring/importing on CO₂ emissions in French and Swedish firms. They also find that offshoring reduces the firm-level emission intensity. Cole et al. (2021) use propensity score matching to show that Japanese firms that start outsourcing production abroad reduce CO₂ emission intensities. Cherniwchan (2017) does not observe actual

⁵Albeit, the standard errors suggest that there is uncertainty about whether offshoring substantially increases global emissions relative to Danish emissions.

⁶Bombardini and Li. (2020) provide evidence for comparative advantage playing a role in deteriorating environmental outcomes in China (as measured by child mortality) in response to their recent export expansion.

plant-level offshoring but documents that NAFTA reduced emissions of local pollutants from manufacturing firms in the U.S. and that this is partly driven by intermediate input tariff reductions.⁷ As for the effects of import competition, Gutiérrez and Teshima (2018) find that increased imports triggered by tariff reductions in Mexico caused firms to increase their energy efficiency, and Choi et al. (2025) show that a reduction in US trade policy uncertainty against China reduced emissions of a local pollutant (particulate matter) from U.S. firms.

In summary, the existing literature has considered different mechanisms while independently focusing on different pollutants. We examine these channels comprehensively using a common measure of a global pollutant with a joint identification strategy. We note that several of the separate pieces of our analysis are not novel in their own right.⁸ Instead, we bring several previously identified ideas and methods together and establish in a coherent empirical framework three important results that have not previously been documented. First, we provide evidence for the effects of import competition and offshoring on firms' emission intensities, output, emissions, energy efficiency, use of electricity, and labor productivity with a joint instrumental variable identification strategy. Second, while import competition has almost no effect on the aggregate emission intensity in the manufacturing sector, by shifting out domestic manufacturing production of intermediate inputs while at the same time increasing output, offshoring has played a major role in reducing the domestic manufacturing emission intensity with relatively modest effects on global emissions. On the other hand, third, Chinese import competition substantially increases global emissions while having almost no effect on the aggregate manufacturing emission intensity.

The rest of the paper is structured as follows. Section 2 describes the firm level data for carbon emissions and international trade transactions. Section 3 outlines the firm-level empirical methodology including our identification strategy. Section 4 presents the esti-

⁷Related to this, Martin (2011) finds that trade reform in India increased firm-level fuel efficiency

⁸For example, Aghion et al. (2024) highlights the importance of accounting for offshoring and import competition jointly but does not consider emissions as an outcome. Dussaux, Vona, and Dechezleprêtre (2023) and Akerman, Forslid, and Prane (2024) also estimate the causal effect of offshoring or importing on firm-level emission intensities using identification approaches from the trade literature (Hummels et al. 2014), but they do not consider the productivity effects of offshoring and reallocation across firms nor do they consider the impact of import competition in final goods. Najjar and Cherniwchan (2021) and Ahmadi, Yamazaki, and Kabore (2022) use point estimates from reduced form regressions to reconstitute terms in a decomposition so as to compute counterfactual changes in sector-level emission intensity, but they consider implications of climate policy rather than effects of offshoring and import competition.

mation results. Section 5 reports decomposition as well as counterfactual decomposition evidence for the aggregate emission intensity in the manufacturing sector as well as leakage rate calculations. Section 6 concludes.

2 Data and Descriptive Statistics

We rely on firm-level data for 1995-2016 from administrative registers in Statistics Denmark. Firms are identified by a unique identifier in the FirmStat register that holds information about basic variables such as industry affiliation (NACE classification), sales, value added, and employment. The firm identifier enables us to merge on two main sources of additional data: firm-level carbon emissions and detailed records of trade flows.

Our measure of firm-level emissions is based on the Industrial Energy Use Survey conducted by Statistics Denmark. The survey covers the universe of manufacturing firms with 20 or more employees, and the firms included capture around 90 percent of total energy use in manufacturing. The survey has a high response rate of 98-99 percent and asks firms about their use of an exhaustive list of energy goods, including oil, coal, and electricity, all measured in gigajoules (GJ). The survey is typically conducted biannually, with the first year being 1995 and the last 2016.⁹ We convert energy in gigajoules to emissions of CO₂ in tonnes using energy-source specific 'emission coefficients'. This approach follows from basic physics, according to which the amount of carbon dioxide emitted from burning, for example, a ton of coal is approximately fixed (Metcalf 2019).¹⁰ See section A.2 for a more detailed description of how we construct the firm-level emissions data. In addition to firm-level Danish emissions, we also use country-product-year specific emission intensities from Exiobase to calculate emissions embodied in import flows (see the results section for more details).

We use the Foreign Trade Statistics Register to measure offshoring and Chinese import competition. Considering HS6 products, we observe imports at the firm-product-origin-year unit of observation. Based on this data, we measure narrow offshoring from all source

⁹The complete list of years we use is 1995, 1996, 1997, 1999, 2001, 2003, 2005, 2007, 2009, 2012, 2014, and 2016.

¹⁰For example Dussaux, Vona, and Dechezleprêtre (2023), Akerman, Forslid, and Prane (2024), Marin and Vona (2021) and Wagner et al. (2020) use a similar approach to measure firm-level carbon emissions for firms in several European countries.

countries following Hummels et al. (2014) and Hummels, Munch, and Xiang (2018). We include all source countries in our offshoring measure as Danish firms rely on suppliers from a wide range of (mainly European) source countries. China, on the other hand, accounts for less than three percent of imported intermediate inputs (see Appendix Table A4).¹¹

Narrow offshoring is intended to capture intermediate inputs that the firm could produce in-house but has decided to import from abroad. Therefore, this measure excludes from the firm's total imports of inputs all imported (HS4) products that the firm never produced or sold, domestically or abroad, during the sample window. Offshoring for firm i in year t is then the total value of the remaining imported inputs, that is

$$\text{OFF}_{it} = \sum_{c,k \in \Omega_i} m_{ckit}, \quad (1)$$

where m_{ckit} is the value of firm i 's import of input k from origin country c at time t , and Ω_i is firm i 's set of country-product input combinations that it could potentially produce in-house. This set is defined from both firm-level production (the PRODCOM register) and exports (Foreign Trade Statistics) data at the HS4 product level.

The Data Appendix shows that offshoring of a particular firm is concentrated in just a few origin-products and that the number of firms for the median offshored origin-product is 1. This means that a supply shock to a particular origin-product affects Danish firms idiosyncratically.

Import competition has traditionally been measured at the industry level (e.g., Bernard, Jensen, and Schott 2006), but our detailed firm-product level trade data allows us to account for the fact that firms within industries may be exposed to import competition to different degrees depending on their product mix. We follow Aghion et al. (2024) and Bernard et al. (2025) and define firm-specific Chinese import penetration as follows:

$$\text{CIP}_{it} = \sum_{k \in \Omega_{i,t_0}^{PROD}} s_{ki,t_0} \frac{M_{kt}^{CN} - M_{ikt}^{CN}}{M_{kt} - M_{ikt} + D_{kt}}, \quad \sum_{k \in \Omega_{i,t_0}^{PROD}} s_{ki,t_0} = 1, \quad (2)$$

where M_{kt}^{CN} and M_{kt} are the values of imports from China and all countries for HS6 product

¹¹In Section 4.3 we examine the impact of offshoring to a particular set of countries.

k at time t , respectively. From these we subtract firm i 's own imports, M_{ikt}^{CN} and M_{ikt} , since these trade flows will not increase competition for firm i (all variables constructed using the Foreign Trade Statistics Register). D_{kt} is total domestic production of product k by Danish firms at time t (using from the PRODCOM register). In total, $\frac{M_{kt}^{CN} - M_{ikt}^{CN}}{M_{kt} - M_{ikt} + D_{kt}}$ measures China's share of the Danish market for each product k and year t .¹²

That is, import competition for firm i is defined as the weighted average of Chinese import penetration in the set of firm i 's produced products in the presample year, Ω_{i,t_0}^{PROD} . The weights, s_{ki,t_0} , are defined as the shares of product k in firm i 's total domestic production over the presample period. The presample period is defined as the first two years the firm is observed in PRODCOM (averaging over two years). This definition holds the pre-sample product mix constant to measure how firms are subsequently affected by surges in imports from China.

In line with the substantial "China shock" literature pioneered by Autor, Dorn, and Hanson (2013) we focus on import competition from China in our preferred specification because Denmark has also seen a substantial increase in imports of products from China. Furthermore, China is outside EU regulations making it particularly policy relevant regarding concerns of carbon leakage and the effects of border adjustment mechanisms.

China's share in total Danish manufacturing imports surged from around 2% in 1995 to 10% in 2016 (see Appendix Figure D1). Furthermore, the share of Danish spending on Chinese goods increased from about 2% in 2000 to about 4.5% in 2007, and 5.7% in 2016.¹³ Overall, this rise in imports from China is similar to import patterns in the United States which had substantial effects on production (Autor, Dorn, and Hanson 2013).¹⁴

Finally, from an identification point of view, Denmark is too small to affect China's comparative advantage, which allows us to construct a plausibly valid instrumental variable.¹⁵

¹²Aghion et al. (2024) use China's import share instead presumably because they do not observe D_{kt} .

¹³For details see Appendix A.5.

¹⁴Autor, Dorn, and Hanson (2013) consider descriptive evidence for imports into other high income economies including Denmark and they state: "Like the United States, these countries experienced a dramatic increase in imports from China between 1991 and 2007..." Furthermore, they report "The share of total US spending on Chinese goods rose from 0.6 percent in 1991 to 4.6 percent in 2007 (Figure 1), with an inflection point in 2001...". Similar to the Danish import share, the U.S. import share in 2001 was about 2%. We do not observe expenditures in 1990, but note that the overall Chinese import share in Denmark was very low.

¹⁵We examine the role of import competition from other countries in Section 4.3.

To construct our manufacturing firm sample, we require all firm-years to have positive emissions, more than twenty employees, more than 300.000 DKK offshoring (following Hummels et al. (2014)), positive sales, positive world export supply (WES) instrument, and defined Chinese export supply (CES) instrument.¹⁶ The resulting number of firm-years is 11,591 across 2019 unique firms, and they cover most of the total carbon emissions (73 percent) and economic activity (roughly two-thirds of sales and value added) in the manufacturing sector (see Table A1 in Appendix A.1).

Table 1 reports key summary statistics for the sample, documenting, for example, that there is considerable within-firm time variation in key variables such as the emission intensity and offshoring. The table also shows that the average firm emits 5,148 tons of CO₂. In addition, Appendix Table A5 documents substantial variation in Chinese import competition across industries as well as across firms within industries.

Table 1: Summary statistics

	Obs.	Nr. Firms	Absolute		Logs		
			Average	Median	Average	St. dev.	Dev. from firm mean
CO ₂	11,591	2,019	5,147	243.88	5.78	1.85	0.41
CO ₂ intensity	11,591	2,019	0.01	0.00	-6.08	1.46	0.45
Offshoring	11,591	2,019	86,480	17,396	9.77	1.72	0.49
Chinese Import Pen.	11,591	2,019	0.05	0.01	-	-	-
Sales	11,591	2,019	453,503	120,310	11.86	1.20	0.23
Value-added	11,591	2,019	134,245	40,326	10.78	1.15	0.25
Employment	11,591	2,019	186.65	72.09	4.45	1.03	0.19

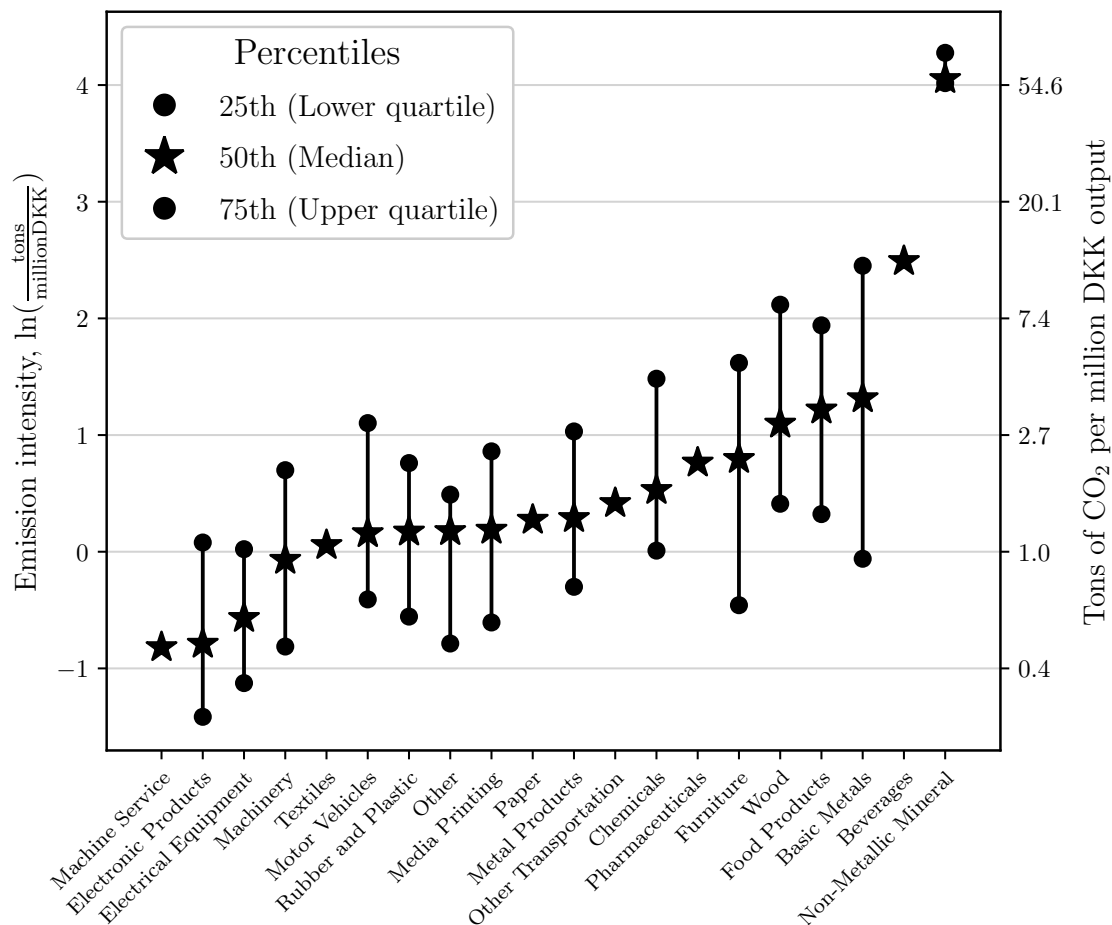
Notes: Monetary variables are measured in thousand DKK, emissions are measured in tons, and employment is measured in annual full-time equivalents. "Dev. from firm mean" denotes the average log-point deviation for any value from its across-year firm mean. Monetary variables are deflated to the year 2015 using the consumer price index.

We now examine the extent of heterogeneity in carbon emission intensities across firms within industries in a single year (2016). Figure 1 shows large differences in median emission intensities across two-digit NACE industries, with Non-Metallic Minerals standing out as the clearly most emission-intensive industry. It is also evident that there is vast heterogeneity across firms within most industries, emphasizing the need to account for this dimension of variation. The average ratio between the 90th and 10th percentile is 382. This means

¹⁶Both instruments will be defined precisely below.

that in the average industry, the firm at the 90th percentile of CO₂ intensity emits 382 times more than the firm at the 10th percentile whenever they produce the same amount of output.¹⁷ Lyubich, Shapiro, and Walker (2018) and Wagner et al. (2020) document similar heterogeneity in firm-level emissions across firms within industries in the U.S. and several European countries.

Figure 1: Heterogeneity in emission intensities across and within industries



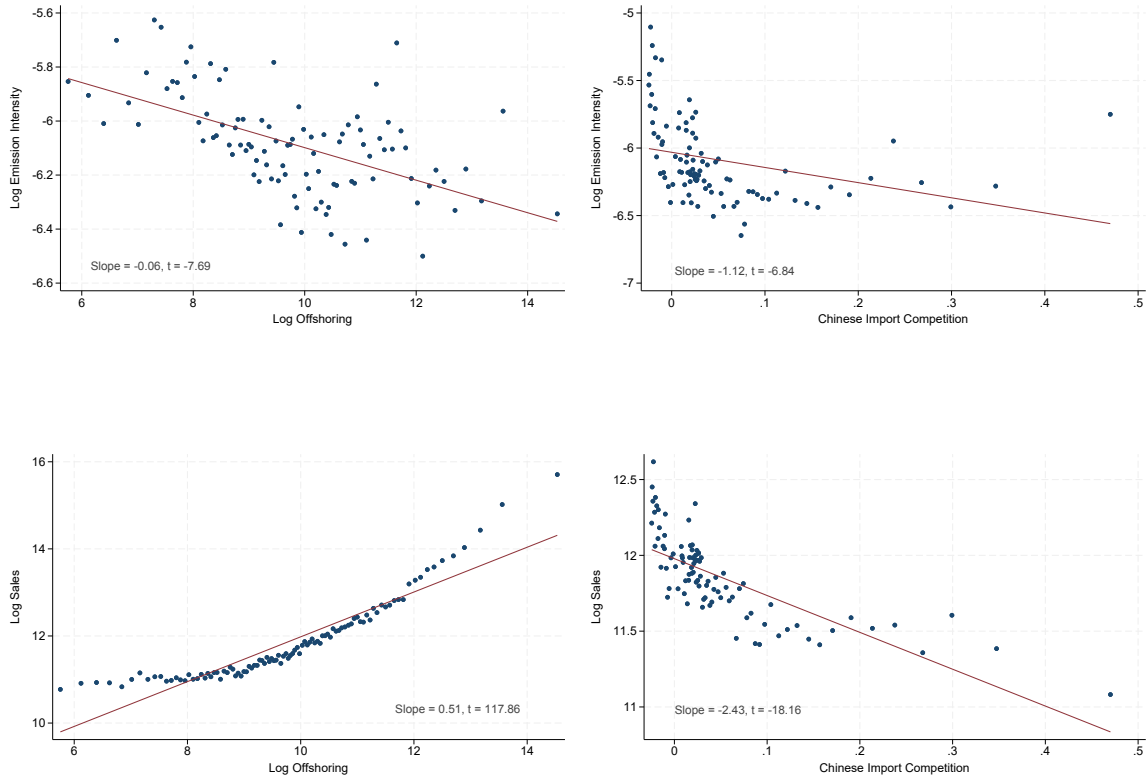
Notes: Firm-level emission intensities are calculated as $\log(\frac{\text{CO}_2}{\text{Output}})$ with CO₂ emissions measured in tons and output measured in million DKK. Some industries have been removed due to too few observations.

We take a first look at simple correlation patterns between our main variables of interest in the raw firm data. The two top panels of Figure 3 show that emission intensities correlate negatively with firm-level offshoring and with firm-level Chinese import competition. This

¹⁷The heterogeneity persists when one changes the industry classification to even the 6-digit level; the average 90/10 percentile ratio is 248.

suggests that relocation of production processes abroad or exposure to increased import competition reduces domestic emissions for a given level of output. The bottom left panel shows a clear positive correlation between offshoring and sales suggesting that offshoring may bring productivity gains for firms. Finally, the bottom right panel documents that sales correlate negatively with Chinese import competition. This has been documented by the literature previously and is consistent with the view that Chinese import exposure acts like a negative demand shock.¹⁸

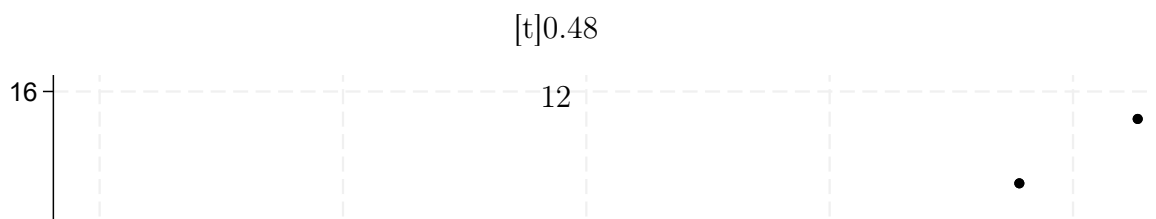
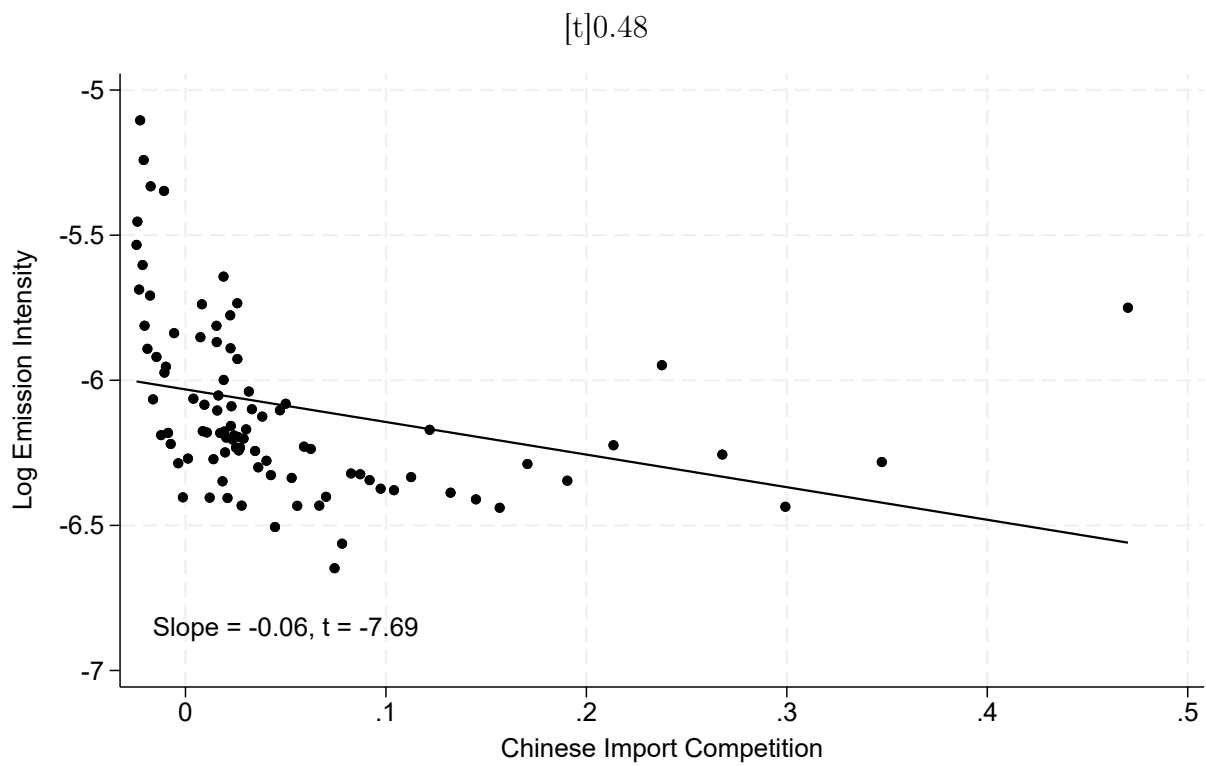
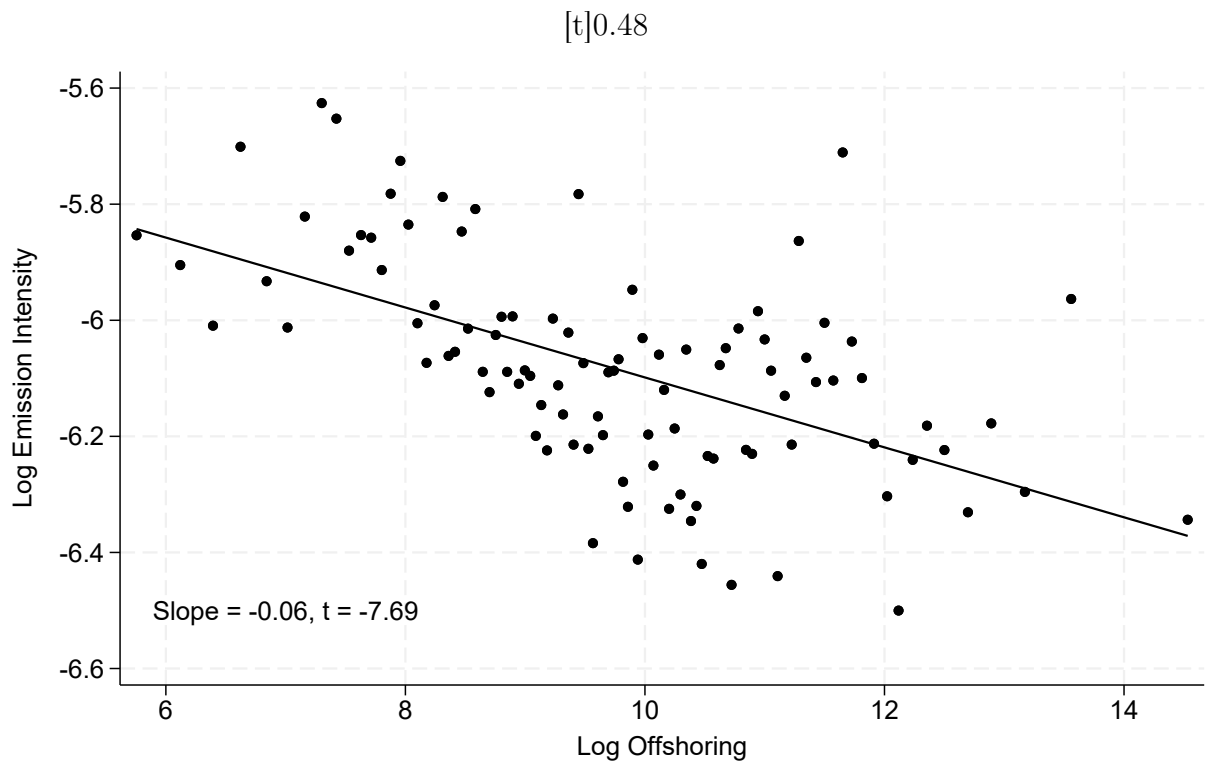
Figure 2: Scatterplots of sales and emission intensities against trade variables



Notes: The four panels show binned scatter plots. The regression lines are based on all underlying observations ($N = 11591$) and controls for year fixed effects but not firm fixed effects. The legends show information about the slope parameter of the regression lines.

¹⁸Appendix Figure D2 illustrates a significant positive relationship between log emissions and log sales. We therefore expect that firm-level carbon emissions should correlate negatively with Chinese import competition. Offshoring is positively correlated with sales but negatively correlated with emission intensities. Thus the relationship between offshoring and emissions is a priori unclear.

Figure 3: Scatterplots of sales and emission intensities against trade variables



In sum, the data patterns provided in this section document substantial cross-firm variation in offshoring and Chinese import competition. However, they do not account for the fact that offshoring and Chinese import competition may be endogenous to firm-level emissions. We will deal with this challenge in the empirical analysis below.

3 Empirical Approach

This section outlines our approach to estimate the impact of our two globalization variables on firm-level (log) emission intensities and firm-level (log) sales. We start with a brief discussion of existing empirical results to motivate our approach. Then, we specify our estimating equations and, relying on existing literature, we define our instrumental variables for offshoring and Chinese import competition.

3.1 Background

Recent literature emphasizes the advantage of decomposing aggregate emission intensities, total emissions divided by output, into firms' market shares and emission intensities (Copeland, Shapiro, and Taylor 2022; Cherniwchan, Copeland, and Taylor 2017). These decompositions highlight the importance of the reallocation of output across heterogeneous firms and changes in firms' emission intensities due to adjustments in production to explain the overall emission content in aggregate output.¹⁹ However, based on the existing literature that emphasizes the effects of Chinese import competition and offshoring on firms' scale of production and productivity, it is not clear what to expect in terms of firms environmental performance.

The empirical trade literature disentangles import shocks into input supply (offshoring) and output (competition) components to jointly examine their effects on output and innovation (Aghion et al. 2024; Mion and Zhu 2013). According to this literature, disentangling the effects is important, because the competition and offshoring components are both part of import shocks but they potentially have opposing effects on firm scale, innovation, and

¹⁹In theory, Cherniwchan, Copeland, and Taylor (2017) further decompose firms' emission intensities into tasks that vary in their emission intensity. Unfortunately, we do not observe tasks and their emission intensity.

efficiency. Therefore, we expect that import competition and offshoring also may have distinct effects on firms' environmental performance, because a firm's emissions are simply determined by the firm's level of production and its emission intensity.

The argument that import competition and offshoring may have opposing effects rests on the existing literature. Globalization leads to a reallocation of market shares, and import competition in particular tends to have a negative effect on economic activity of exposed firms and local labor markets (Melitz 2003; Bernard, Jensen, and Schott 2006; Autor, Dorn, and Hanson 2013, 2016; Dauth, Findeisen, and Suedekum 2014). The literature provides mixed evidence for the effects of import competition on innovation and productivity (Bloom, Draca, and Van Reenen 2016; Autor et al. 2020; Pavcnik 2002). Offshoring, on the other hand, has the potential to improve efficiency, reduce costs, and increase sales (Grossman and Rossi-Hansberg 2008; Amiti and Konings 2007).

Based on this evidence, import competition and offshoring may have similar or opposing effects on emission intensities.²⁰ Therefore, empirical evidence that does not disentangle the effects of import competition and offshoring leads to inconclusive results regarding the effects of importing on emission intensities. Consequently, the following subsections develop an empirical approach that jointly identifies the effects of import competition and offshoring on firm level output and emission intensities.

3.2 Regression Specifications

This subsection explains the empirical specifications we estimate to examine the effects of offshoring and Chinese import competition as defined in (1) and (2) on firm-level sales and emission intensities.

We relate log emission intensities to offshoring and Chinese import competition by estimating the following regression equation:

$$\ln \frac{\text{Emissions}_{it}}{\text{Sales}_{it}} = \beta \ln \text{OFF}_{it} + \lambda \text{CIP}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (3)$$

²⁰Relatedly, Cole, Elliott, and Okubo (2014) develops a model where firms may reduce emissions by either investing in abatement or by offshoring. We abstract from this firm level choice by examining how exogenous changes in offshoring opportunities affect firms.

where i indexes firms, t indexes years, α_i is a firm fixed effect, γ_t is a year fixed effect, and ε_{it} is an error term. The two parameters of interest are β , the elasticity of the emission intensity with respect to offshoring, and λ , the semi-elasticity of the emission intensity with respect to import competition from China. Whether these elasticities are positive or negative is an empirical question as discussed above.

If firms offshore intermediate inputs and keep output unchanged, then, all else equal, we expect $\beta < 0$, because in that case domestic production decreases relative to the total output produced and sold. However, this is not a forgone conclusion if, for example, the emission intensity is measured in terms of value added and firms offshore the least emission intensive inputs. In that case, offshoring may actually increase emissions per dollar of value added created in the firm (Cherniwchan, Copeland, and Taylor 2017).²¹

Import competition reduces emission intensities if innovation activities increase in response to competition leading to productivity improvements such that production requires lower amounts of inputs and in particular lower amounts of energy per unit of output (Bloom, Draca, and Van Reenen 2015). Furthermore, firms may switch to cleaner energy sources. In that case we expect $\lambda < 0$. This is also not a foregone conclusion. Firms may switch to dirtier energy sources in response to import competition. In addition, while firms' output may decrease in response to import competition, firms' energy needs to keep the lights on etc. may not be as flexible. In that case, we expect that a decrease in output associated with a relatively constant energy demand raises emission intensities.

To examine if offshoring and import competition affect firm-level sales we estimate

$$\ln \text{Sales}_{it} = \theta \ln \text{OFF}_{it} + \delta \text{CIP}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (4)$$

where again α_i is a firm fixed effect and γ_t is a year fixed effect. The parameters of interest are θ and δ , the elasticity of firm-level sales with respect to offshoring and the semi-elasticity of firm-level sales with respect to Chinese import competition.

If offshoring causes firms to become more productive and larger, then we should expect sales to rise and the offshoring elasticity θ is positive (Grossman and Rossi-Hansberg 2008).

²¹We investigate this in a robustness exercise defining the emission intensity relative to value added.

If exposure to Chinese import competition is akin to a negative demand shock, then we expect firm-level sales to fall and the import competition semi-elasticity δ is negative.

Note that the elasticities θ and γ also represent the effects of import competition and offshoring on log market shares. The reason is that the denominator of market shares (market size) is identical across all firms in a year and is log separable from sales in the log market share. As a consequence, it is fully absorbed by the year fixed effect γ_t . We will take advantage of this when we examine the effect of import competition and offshoring on the decomposition of the aggregate emission intensity.

For completeness, we also relate log emissions to offshoring and Chinese import competition according to the relationship:

$$\ln \text{Emissions}_{it} = \rho \ln \text{OFF}_{it} + \mu \text{CIP}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad (5)$$

where again α_i is a firm fixed effect and γ_t is a year fixed effect. The parameters of interest are ρ and μ , the elasticity of firm-level emissions with respect to offshoring and the semi-elasticity of emissions with respect to Chinese import competition. Note that firm level emissions are log separable into firm-level sales and emission intensities (emissions per unit of sales) such that $\ln \text{Emissions}_{it} = \ln \frac{\text{Emissions}_{it}}{\text{Sales}_{it}} \times \ln \text{Sales}_{it}$. Therefore, the estimates elasticities based on (3) and (4) simply add up to the elasticities in (5): $\rho = \theta + \beta$ and $\mu = \delta + \lambda$. In other words, the offshoring and import competition mechanisms that determine sales and emission intensities captured in specifications (3) and (4) add up to the effects on the firms' emission output.

We estimate (3), (4), and (5) using a two-stage least squares instrumental variable estimator to examine the effects of offshoring and Chinese import competition on emission intensities, sales, and emissions. For comparison, we will also estimate specifications including only one of the two endogenous variables of interest.

3.3 Identification

We now explain how we define valid instrumental variables for offshoring and Chinese import competition to estimate equations (3), (4), and (5). Offshoring may be endogenous to

firm performance affecting both sales and emission intensity if, for example, unobserved productivity shocks induce firms to offshore and sell more and to become cleaner (productivity correlates with energy efficiency, Copeland, Shapiro, and Taylor (2022)). Chinese import competition may also be endogenous to firm performance if, for example, Chinese firms' activities tend to increase more in industries with poorly performing domestic firms. As is standard, we require that the instrumental variables that address these sources of potential endogeneity meet the exclusion restriction and predict the endogenous variables offshoring and import competition.

To generate exogenous variation in offshoring, we employ a 2SLS procedure applying a shift-share Bartik instrument similar to the World Export Supply (*WES*) instrument developed in Hummels et al. (2014). Formally, the instrument is calculated as

$$WES_{it} = \sum_{c,k \in \Omega_{i,t_0}^{IMP}} s_{cki,t_0} \times X_{ckt}, \quad \sum_{c,k \in \Omega_{i,t_0}^{IMP}} s_{cki,t_0} = 1, \quad (6)$$

where s_{cki,t_0} (the "shares") denote product k from country c 's share of firm i 's total imports in the pre-sample years t_0 . That is, Ω_{i,t_0}^{IMP} is firm i 's full set of country-product combinations that it imported in the presample year, t_0 . X_{ckt} (the "shift") is exports from country c of product k to the rest of the world at time t . To avoid confounding factors, we exclude both Denmark and China when defining *WES*. Products are measured at the HS6 level. The pre-sample years are firm-specific and defined as the two immediate years before the firm first reports emissions and hence enters the sample. For the instrument to be defined and positive, at least one of the origin-products (source country and HS6 product code combination) imported in Denmark must simultaneously be exported to a different country (excluding China).

The instrument exploits that Danish importers source different products from different countries. For example, a Danish cement producer importing cement bricks from Germany will likely offshore more, in other words, import more cement bricks, if producers of cement bricks in Germany can produce at lower costs. At the same time, if German producers of cement bricks produce with lower costs it will show up in the global trade data as increased exports of cement bricks from Germany to the rest of the world. This is the "shift" part

of the instrument. Whether a firm is exposed indirectly to productivity shocks of German cement brick producers depends on whether the firm is already tied to those producers. Therefore, a measure of this exposure is whether the firm initially imports cement bricks from Germany. This is the idea behind the "share" part of the instrument.

The necessary and sufficient conditions for consistency of the 2SLS estimator rely on of Borusyak, Hull, and Jaravel (2021). The central condition is that the shocks - in our setting export flows of a particular product from a particular origin to the world excluding Denmark and China - are as-good-as-randomly assigned from the perspective of the Danish firm. Appendix A shows that any given firm's imports are concentrated on just a few products, the importing behavior is relatively stable in terms of origin-products imported and that different firms import very different origin-products. The two latter insights help ensure relevance and validity of the shift-share instrument, respectively.

We address possible endogeneity of Chinese import competition following the existing literature by defining an instrumental variable that captures China's improving comparative advantage as reflected in growing export supply to the rest of the world (Autor, Dorn, and Hanson 2013). The Chinese import competition variable as defined in equation (2) measures China's market shares in Denmark. As a consequence, we define the instrument as a shift share variable, where the shifts are Chinese market shares in all third countries. To be specific, the Chinese export supply share instrument is

$$CESS_{it} = \sum_{k \in \Omega_{i,t_0}^{PROD}} s_{ki,t_0} \frac{CES_{kt}}{ES_{kt}}, \quad \sum_{k \in \Omega_{i,t_0}^{PROD}} s_{ki,t_0} = 1, \quad (7)$$

where CES_{kt} is China's total supply of product k to the entire world, minus exports to Denmark, in period t , and ES_{kt} is all countries' (excluding Denmark's but including China's) exports of product k to the entire world. That is, $\frac{CES_{kt}}{ES_{kt}}$ is China's share of the world market in product k . These market shares are then weighted by shares s_{ki,t_0} of product k in firm j 's total domestic production in the presample year. The presample shares are identical to those that enter the definition of CIP_{it} in equation (2).

The identifying assumption is that the market shares of China in the rest of the world only affect Danish firms through its effect on the Chinese import competition measure. When

Chinese exporters become more productive, exports to the rest of the world increases relative to other countries' exports to these countries. At the same time, more productive exporters in China should lead to a larger market share for Chinese firms in Denmark. We will provide evidence that the instruments appropriately predict the endogenous variables based on first stage regressions.

4 Results

In this section we first present first-stage results from our 2SLS procedure. We then turn to our main second-stage results from estimating equations (3), (4), and (5). Finally, we report results from a number of extensions and robustness exercises.

4.1 First-Stage Results

Table 2 provides results from the first-stage regressions. Column (1) shows that our world export supply instrument for offshoring has a positive and significant effect on offshoring. Therefore, as anticipated in Section 3.3, if the cost advantage of Danish firms' foreign suppliers improves in global markets, then Danish firms source a greater amount of inputs they could have produced in-house from their foreign suppliers. Furthermore, F-statistics confirm that world export supply is, by conventional standards, a strong instrument for Danish offshoring activities.

Column (2) shows that the Chinese export supply share instrument predicts the firm-level Chinese import competition variable as expected. As China's comparative advantage improves in world markets, Danish firms experience a greater amount of import competition in final product markets. Again, the F-statistics provide evidence that the Chinese export supply share variable is a strong instrument.

Finally, columns (3) and (4) relate each endogenous variable to both instrumental variables. Column (3) shows that world export supply significantly explains offshoring with almost the same coefficient as in column (1), while the effect of the Chinese export supply share is insignificant. Column (4) shows the reverse. The Chinese export supply share significantly explains Chinese import competition, while the world export supply does not.

In summary, the first-stage results show that the world export supply and Chinese export supply share instruments capture distinct global market forces that affect Danish firms via different mechanisms. Importantly, the first-stage results also provide evidence that our instruments allow us to separately identify the effects of offshoring and Chinese import competition on firm outcomes in specifications (3), (4), and any other specification we estimate throughout the paper. Therefore, based on these first-stage regressions, we turn to discussing second-stage results of specifications (3) and (4) in subsection 4.2 and robustness checks and extensions in subsection 4.3.

Table 2: First Stage Results

	ln Offshoring (1)	CIP (2)	ln Offshoring (3)	CIP (4)
ln WES	0.1402*** (0.0384)		0.1344*** (0.0388)	0.0005 (0.0022)
CESS		0.4105*** (0.0220)	-0.3222 (0.2172)	0.4109*** (0.0221)
F-stat (WES)	13.4		12.0	0.1
F-stat (CESS)		347.4	2.2	345.6
Observations	11,591	11,591	11,591	11,591

Notes: All columns show first stage regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

4.2 Second-Stage Results

We now turn to the effects of offshoring and Chinese import competition on firm-level emission intensities defined as CO₂ emissions divided by sales (columns (1)-(3)), sales (columns (4)-(6)), and firm-level emissions (columns (7)-(9)) in Table 3.²²

The first three columns show that offshoring and Chinese import competition reduce firm-level emission intensities whether entered separately or jointly. Furthermore, the standard errors imply that the effects of offshoring and Chinese import competition on emission

²²Appendix Table C1 shows the corresponding OLS results.

intensities are significantly different from zero.²³

The estimated offshoring elasticity in the preferred joint specification of column (1) implies that a one percent increase in offshoring leads to a 0.67% fall in the emission intensity, all else equal. We verify that the elasticity estimate of 0.67 is economically meaningful for firms' environmental performance by quantifying the implication of observed five-year changes in offshoring activity on a firm's emission intensity. At the 10th percentile, a firm's five-year log change in offshoring is about -0.74. Based on an elasticity of -.67, this fall in offshoring corresponds to a 64.2% higher emission intensity.²⁴ By contrast, the log change of the 90th percentile firm is about 1.35, which leads to a 59.5% lower emission intensity.²⁵ We conclude that the elasticity estimate leads to economically relevant changes in firms' emission intensity that vary substantially across the distribution of observed offshoring activities.

Table 3: Firm-level Effects of Offshoring and Chinese Import Competition

	ln CO ₂ /Sales			ln Sales			ln CO ₂		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ln Off	-0.674** (0.282)	-0.574** (0.249)		0.617*** (0.168)	0.687*** (0.175)		-0.057 (0.233)	0.113 (0.210)	
CIP	-2.003*** (0.712)		-1.316** (0.556)	-1.409*** (0.366)		-2.038*** (0.301)	-3.412*** (0.591)		-3.354*** (0.530)
Obs.	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591

Notes: All columns show second-stage regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The estimated semi-elasticity for Chinese import competition in Table 3 column (1) implies that when Chinese imports rise by one percentage point, the emission intensity falls by 2%, all else equal. Evaluated at the mean value of the Chinese import penetration variable from Table 1, this corresponds to an elasticity of -0.1. In other words, compared with the offshoring elasticity, Chinese import penetration has a weaker impact on emission intensities

²³We report robust standard errors. Abadie et al. (2023) show that clustering can lead to "severely inflated" standard errors, especially when the sample is a large fraction of the population (our dataset includes the universe of firms with at least 20 employees) and when there is variation in treatment within clusters (which applies to firm-level clustering in our context). When we cluster at the firm level, the only meaningful change is that the standard errors of the effects of offshoring and import competition on the emission intensity increase slightly, and we have to drop a star.

²⁴To be precise, we compute $[\exp(-.67 \times -0.74) - 1] \times 100 \approx 64.2$.

²⁵We compute $[\exp(-.67 \times 1.35) - 1] \times 100 \approx -59.5$.

of Danish firms.

We illustrate the economic importance of this semi-elasticity estimate for firms' environmental performance by considering the effect of greater Chinese import competition on Danish firms' emission intensity in our data sample over five-year periods. The 10th percentile firm has a modest drop of 1.1 percentage points in Chinese import competition, which approximately corresponds to a 2.2% rise in emission intensities.²⁶ The 90th percentile firm sees a substantial increase of 7.6 percentage points in Chinese import competition, which corresponds to a 14.1% fall in emission intensities.²⁷ We conclude that the effect of import competition on firms' emission intensity is more modest than the effect of firms' offshoring activity. Nevertheless, similar to the effects of offshoring on emission intensities, the effects vary substantially across the firm-level distribution of Chinese import competition.

Columns (2) and (3) report estimates for the effects of offshoring and import competition on emission intensities when we do not jointly account for both globalization mechanisms. The estimates from the joint and non-joint specifications are fairly similar. This is another indication for the validity and strength of our instruments and their usefulness to separately identify the effects of import competition and offshoring. That said, based on the recent existing literature, the preferred estimates are from the specifications reported in column (1) (Aghion et al. 2024).

We next examine the effects of import competition and offshoring on firms' sales. Table 3 columns (4-6) show the results. Across all columns, offshoring raises firms' sales while Chinese import competition reduces sales. Furthermore, all the estimates are statistically significant.

According to our preferred specification in column (4), an exogenous increase in offshoring causes firm sales to rise with an elasticity of 0.62. Thus, offshoring may induce firms to become more productive.²⁸ Again, the heterogeneity in offshoring changes across firms means that sales respond very differently. A firm at the 10th percentile sees sales falling by about 36.7% as a result of changes in offshoring over a five-year period, while sales of the 90th

²⁶We compute $[\exp(-2.003 \times -0.011) - 1] \times 100 \approx 2.23$.

²⁷We compute $[\exp(-2.003 \times 0.076) - 1] \times 100 \approx -14.12$.

²⁸Appendix Table C2 shows that offshoring has a positive effect on labor productivity as measured by value added per worker.

percentile firm rise by about 130%.²⁹

The estimated semi-elasticity for Chinese import competition reported in column (4) means that when import competition increases by one percentage point, sales fall by 1.4%, all else equal. In the distribution of five-year changes in Chinese import competition, the 10th percentile firm experiences a 1.5% rise in sales, while the 90th percentile firm sees a substantial fall in sales of 10.7%.

For comparison, columns (5) and (6) report estimates from specifications that include import competition and offshoring individually. The estimates are again comparable to the estimate from the joint specification in column (4).

Finally, columns (7-9) show that offshoring does not affect firm-level emissions. This is a reflection of the results in columns (3) and (6); offshoring reduces emissions per unit of output, but output rises. In total, this leaves firm-level emissions unchanged. By contrast, Chinese import competition reduces both emissions per unit of sales and sales volumes. Consequently, firm-level emissions fall in response to Chinese import competition. According to column (7), a one percentage point increase in Chinese import competition reduces emissions by 3.4%. The estimate in column (9) shows that excluding offshoring from the specification does not affect this conclusion.

To summarize our firm-level evidence, we find that offshoring and import competition have significant but distinct effects on firm-level emission intensities, sales, and emissions. Offshoring leads to cleaner production at the firm level but also raises output such that firm-level emissions do not change significantly. On the other hand, Chinese import competition reduces firm-level emissions both through lower emission intensities and lower sales.

4.3 Extensions and Robustness

In this section, we explore mechanisms, heterogeneity in effects, alternative definitions of offshoring and import competition, and value added instead of sales in the firm-level emission intensity variable. For all of these extensions and robustness exercises, we focus on our preferred approach that includes both import competition and offshoring in the same specification, and we report 2SLS estimates.

²⁹Similar as above we compute $[\exp(0.617 \times -0.74) - 1] \times 100$ and $[\exp(0.617 \times 1.35) - 1] \times 100$.

We first examine potential mechanisms that explain how firms adjust to increasing globalization. Offshoring reduces the firm-level emission intensity, and one question is whether this is accomplished by changing the firm-level mix of energy inputs. For example, if offshoring moves out heavily energy-consuming production processes that require the use of relatively dirty energy inputs such as coal or fuel, then the share of cleaner energy inputs such as electricity may rise.

We address this question by decomposing the effects of offshoring and import competition on the firm-level emission intensity (CO_2 per unit of sales) reported in Table 3 into an emission factor (CO_2 per GJ) and an (inverse) energy efficiency component (GJ per sales). The emission factor captures energy input substitution as measured by the emission of carbon per unit of energy input, GJ, and the energy efficiency component measures energy per unit of sales. For comparison, Table 4 column (1) reproduces the estimates of the effects of offshoring and import competition on emission intensity from Table 3 column (1). Column (2) reports the estimates of the effect of import competition and offshoring on the emission factor, and column (3) reports the estimates for energy efficiency. We note that the effect of, for example, offshoring on emission intensity is the sum of its effects on the emission factor and on energy efficiency.

The estimates in Table 4 indicate that offshoring lowers the emission factor. By contrast, GJ per sales decreases (energy efficiency improves) in response to offshoring albeit not in a statistically significant way. The negative effect of Chinese import competition on the emission intensity is explained roughly equally by improved energy efficiency and a lower emission factor. Therefore, while offshoring primarily leads to cleaner production by changing energy inputs, import competition reduces emission intensity mainly through improvements in efficiency but also by adjusting towards cleaner energy inputs.

We provide additional evidence regarding energy switching by directly examining whether globalization affects the use of a relatively clean energy input, electricity. Across the columns of Table 5 we examine the effect of import competition and offshoring on firms' use of electricity (column 1), their use of non electricity energy inputs (column 2), their total energy use (column 3), and the share of electricity use relative to their total energy use (column 4).

Table 4: Energy Input Substitution and Energy Efficiency

	ln CO ₂ /Sales	ln CO ₂ /GJ	ln GJ/Sales
	(1)	(2)	(3)
ln Off	-0.674** (0.282)	-0.437** (0.183)	-0.237 (0.192)
CIP	-2.003*** (0.712)	-0.874* (0.471)	-1.129*** (0.419)
Observations	11,591	11,591	11,591

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Column (1) of Table 5 shows that offshoring has a positive impact on the use of energy from electricity but no impact on the use of non-electricity energy inputs (column 2).³⁰ As a consequence, total energy use rises (column 3), and the electricity share in energy inputs rises (column 4), although this effect is not significant.³¹ Chinese import competition lowers both electricity and non-electricity inputs, consistent with the finding that Chinese import competition is akin to a negative demand shock for exposed firms that scales down firm-level activity. The negative effect on non-electricity inputs is greater in magnitude compared to the electricity input, leading to a higher electricity share (column 4).³²

We next examine if our globalization variables affect firms differently depending on whether they are clean or dirty. Our reasoning for this robustness check is straightforward. If a firm is already clean or if clean firms are not subject to offshoring and import competition forces, then we would not expect that either import competition or offshoring leads to even cleaner production. To empirically implement this, we split firms into two groups and label firms as dirty in their production if their initial emission intensities are in the fourth quartile of firm-level emission intensities. We then interact the globalization

³⁰The average electricity share in the sample is 0.47 and the median is 0.44.

³¹A potential concern of using sales in our previous specifications is that firm-level sales capture price instead of output changes. While we cannot rule price adjustments out entirely, the results in Table 4 and Table 5 provide evidence for input adjustments associated with changes in output. Furthermore, Appendix Figure D2 shows that input-based emissions are closely related to sales. We would not expect such a strong relationship if the variation in sales was mostly driven by price variation.

³²As an additional robustness check, we re-estimated our coefficient excluding industries that do not depend as heavily on electricity, such as iron, steel, aluminum, and cement production. The estimates remain similar and are reported in Table C3. However, we now find that offshoring has a significant effect on the electricity share, albeit only at the 10% level.

Table 5: Electricity switching

	ln Electricity (GJ)	ln Non Electricity (GJ)	ln Total GJ	ln Electricity share
	(1)	(2)	(3)	(4)
ln Off	0.571*** (0.216)	-0.062 (0.258)	0.349* (0.183)	0.222 (0.137)
CIP	-2.125*** (0.464)	-3.500*** (0.619)	-2.761*** (0.402)	0.636** (0.297)
Observations	11,514	11,514	11,514	11,514

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

variables with this indicator distinguishing firms that exhibit relatively dirtier production in the initial period.

Table 6 reports the results when we distinguish effect by firms that initially were relatively dirty. Column (1) and (2) show that offshoring has a similar impact on emission intensities and sales for clean and dirty firms as the interaction term for offshoring is insignificant in both columns. By contrast, we find that emission intensities and sales fall by much more in dirty firms than in clean firms when they are hit by Chinese import competition shocks. This may suggest that dirty local firms are more directly exposed to Chinese import competition, potentially because they produce more homogeneous products.

Table 6: Effect Heterogeneity

	ln CO ₂ Intensity	ln Sales
	(1)	(2)
ln Off	-0.635** (0.264)	0.600*** (0.162)
ln Off. $\times$ 4. Quartile of CO ₂ Int.	-0.068 (0.203)	0.172 (0.163)
CIP	-1.405* (0.833)	-0.993** (0.478)
CIP $\times$ 4. Quartile of CO ₂ Int.	-3.358** (1.387)	-1.691* (0.981)
Observations	11,591	11,591

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We now re-examine the definition of our globalization variables. We have so far examined

import competition from China and offshoring to all countries. However, import competition from other low-income countries or offshoring to a particular set of countries may affect the performance of domestic firms in distinct ways. For example, Dauth, Findeisen, and Suedekum (2014) finds that import competition from emerging Eastern European countries plays a dominant role in explaining labor market outcomes in Germany. The share of total imports in Denmark from the new Central and Eastern European EU member countries has risen in a similarly dramatic way as imports from China, see Appendix Figure D1.

In Table 7 we estimate the impact of offshoring to and import competition from Eastern Europe (defined by the new Eastern European EU member countries), both measured at the firm level. The sample size is cut by more than half because many firms do not offshore intermediate inputs to Eastern European countries. The effect of offshoring to Eastern European countries on emission intensities is still negative, but it is now insignificant (columns (1) and (3)). Offshoring to Eastern European countries still has a positive impact on sales (columns (4) and (5)). It is only significant when entered separately, but even in that case the first stage is weak. It is also evident that import competition from Eastern Europe does not have a significant impact on emission intensities or sales (columns (1), (3), (4), and (6)). In summary, while we find that in general offshoring and Chinese import competition have substantial effects on firms' environmental and economic performance (see Table 3), we do not find evidence for such effects due to the enlargement of the European Union with Eastern European economies.

We next examine the importance of using value added instead of sales as a measure of firm-level activity. Table 8 reports results where we use value added as the denominator in the firm-level emission intensity, and we use log value added as the dependent variable rather than log sales. Qualitatively, we obtain similar results, and the estimated elasticities change only marginally. This suggests that the within-firm time variation in sales and value added, driven by offshoring and import competition, is similar.

A potential identification concern may be that unobserved environmental regulation may confound the impact of globalization variables on emission intensities in Danish firms. The most important environmental regulation in the EU is the Emissions Trading System (ETS), which is the world's largest market-based climate policy covering emissions in emission-

Table 7: Eastern European Countries

	ln CO ₂ Intensity			ln Sales		
	(1)	(2)	(3)	(4)	(5)	(6)
ln Off CEEC	-0.462 (0.409)	-0.329 (0.205)		0.405 (0.306)	0.310** (0.154)	
IP CEEC	9.279 (8.173)		1.223 (1.587)	-6.562 (6.001)		0.492 (0.805)
F-stat (log Off.), WES	3.2	4.7		3.2	4.7	
F-stat (log Off.), ESS	11.1			11.1		
F-stat (ImpPen), WES	0.0			0.0		
F-stat (ImpPen), ESS	10.3		10.5	10.3		10.5
Observations	4,932	4,932	4,932	4,932	4,932	4,932

Notes: All columns show 2SLS regressions and include year and firm fixed effects. CEEC refers to the new Central and Eastern European EU member countries. The lower panel shows information on the first-stage regressions: F-statistic for the test of weak instruments. WES denotes the world export supply instrument for offshoring to CEEC, and ESS denotes the export supply share instrument for import competition from CEEC. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Value Added

	ln CO ₂ Int. (VA)	ln Value Added
	(1)	(2)
ln Offshoring	-0.658** (0.308)	0.573*** (0.186)
CIP	-1.692** (0.765)	-1.725*** (0.424)
Observations	11,543	11,543

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

intensive industries.³³ To account for the ETS, we obtain a list of Danish firms covered by the EU ETS from the European Commission³⁴ and define a variable indicating if a firm is part of ETS in a given year. Table 9 shows that our main results in Table 3 are almost unchanged when including this ETS variable as a control.

Table 9: Control for ETS

	ln CO ₂ Intensity			ln Sales			ln CO ₂		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ln Off	-0.673** (0.281)	-0.577** (0.247)		0.616*** (0.168)	0.683*** (0.172)		-0.057 (0.232)	0.106 (0.207)	
CIP	-2.012*** (0.743)		-1.205** (0.564)	-1.398*** (0.382)		-2.137*** (0.307)	-3.411*** (0.616)		-3.342*** (0.539)
ETS	-0.017 (0.119)	0.076 (0.098)	0.195*** (0.060)	0.020 (0.073)	0.085 (0.073)	-0.174*** (0.040)	0.003 (0.098)	0.161* (0.083)	0.021 (0.067)
Obs.	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591

Notes: All columns show second-stage regressions and include year and firm fixed effects. ETS is a dummy variable indicating if a firm in a given year is part of the EU Emissions Trading System. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Finally, Appendix Table C4 shows the implications of adding an export dummy, (log) export volume, and firm-level employment and capital as control variables. One concern may be that the estimated offshoring and Chinese import competition effects also capture the effects of exporting. Existing studies have documented that exporting causes firms to reduce their emission intensity (Forslid, Okubo, and Ullveit-Moe 2018; Rodrigue, Sheng, and Tan 2024b), and that firms that import are also the firms that are engaged in exporting (Bernard et al. 2018). However, Appendix Table C4 shows that our results are robust to the inclusion of either an export dummy or (log) export volume as controls. Controlling for capital and employment, offshoring still has a negative impact on the emission intensity and a positive impact on sales, but the impact of Chinese import competition on sales is muted when employment and capital are kept fixed.

Our firm-level estimates provide clear evidence for the effects of offshoring and Chinese import competition on firm-level emission intensities, sales, and emissions. However, based

³³Recent evidence shows that the introduction of the ETS in 2005 reduced emissions in French firms without affecting sales (Colmer et al. 2025).

³⁴<https://union-registry-data.ec.europa.eu/report/welcome>

on this firm-level evidence alone, it is unclear how these effects translate into changes in manufacturing sector’s aggregate emission intensity or emission leakage. The reason is that sector-wide aggregates depend on the “sum” of firm-level changes in emission intensities and market shares (sales) in response to varying changes in offshoring and Chinese import competition across the firm distribution. Therefore, based on the firm-level evidence, the following section examines the effects of offshoring and Chinese import competition on the manufacturing sector, providing decomposition and counterfactual decomposition evidence.

5 Decomposition Evidence and Carbon Leakage

This section uses the coefficient estimates from the previous sections to examine the role of offshoring and Chinese import competition in the decline of the aggregate manufacturing sector emission intensity. We start by decomposing the average manufacturing sector emission intensity, focusing on the distinct effects of offshoring and Chinese import competition identified by our firm-level evidence: changes of emission intensities within firms and the reallocation of market shares (sales) between firms. We then use our decomposition to provide counterfactual evidence that evaluates the effects of offshoring and Chinese import competition on the manufacturing sector’s emission intensity.

5.1 The Decomposition

Following Foster, Haltiwanger, and Krizan (2001) we define the aggregate emission intensity Φ_t in the manufacturing sector as the market-share weighted sum of firm-level emission intensities

$$\Phi_t = \sum_i s_{it} \phi_{it}, \quad (8)$$

where s_{it} denotes firm i ’s market shares in period t , with $\sum_i s_{it} = 1$, and ϕ_{it} is firm’s emission intensity and.³⁵ Then, the change in the aggregate emission intensity between 1996 and 2016

³⁵Decomposing weighted averages is a common approach in the firm-level productivity decomposition literature, for example, see Melitz and Polanec (2015).

decomposes perfectly into four components of interest,

$$\begin{aligned}
\frac{\Phi_{2016} - \Phi_{1996}}{\Phi_{1996}} &= \underbrace{\frac{\sum_{i \in S} s_{i1996} (\phi_{i2016} - \phi_{i1996})}{\Phi_{1996}}}_{\text{Within firm}} + \underbrace{\frac{\sum_{i \in S} (s_{i2016} - s_{i1996}) (\phi_{i1996} - \Phi_{1996})}{\Phi_{1996}}}_{\text{Between firm}} \\
&+ \underbrace{\frac{\sum_{i \in S} (s_{i2016} - s_{i1996}) (\phi_{i2016} - \phi_{i1996})}{\Phi_{1996}}}_{\text{Cross/interaction}} \\
&+ \underbrace{\frac{\sum_{i \in E} s_{i2016} (\phi_{i2016} - \Phi_{1996}) - \sum_{i \in X} s_{i1996} (\phi_{i1996} - \Phi_{1996})}{\Phi_{1996}}}_{\text{Net Entry = Entry - Exit}} \quad (9)
\end{aligned}$$

The set S refers to firms that exist in both 1996 and 2016, the set E refers to entrants present in 2016 only. The set X refers to firms that are active only in 1996 and exit.

The within, between, and cross components capture the effect of changing emission intensities and market shares on the overall emission intensity. The within effect measures the contribution from changing firm-level emission intensities for a given initial market share, and the between firm effect measures the contribution from changing firm-level market shares for given initial emission intensities. The cross term measures the contribution of firms that see simultaneous adjustments in market shares and emission intensities. Finally, the net entry effect measures the contributions from entering and exiting firms.

Table 10 row 1 reports the overall change in the manufacturing emission intensity and its four components for our sample of firms described in Table 1.³⁶ From 1996 to 2016, the aggregate manufacturing emission intensity declined by about 50%. The within and between firm components account for a decline of 19 and 20 percentage points, respectively. Therefore, they combine to account almost equally for more than three quarters of the total decline in the emission intensity.

The cross term contributes only 1 percentage point to the total decline in the overall emission intensity, and the net entry of firms contributes to the decline in emission intensities by about 13 percentage points, or roughly one quarter of the overall decline in the manufacturing emission intensity.

³⁶Only firms that either exist in the first year, 1995, the last year, 2016, or both are included in the decomposition. That is, firms that enter after the first year and exit before the last year are excluded. This explains why the number of firms in Table 1 is larger than in Table 10.

Table 10: Decomposition of the change in the aggregate emission intensity

	Observations	Total	Within firm	Between firm	Cross	Net entry
Unbalanced panel	1829	-0.50	-0.19	-0.20	0.01	-0.13
Balanced panel	726	-0.55	-0.28	-0.29	0.02	0

Notes: The table shows decompositions of the relative change in the aggregate emission intensity as defined in equation (9). Both rows show decompositions over the entire period from 1996 to 2016. The second row performs the decomposition on a balanced panel where the net entry effect is zero by construction because entering and exiting firms have been removed.

For the remainder of this section, we focus on adjustments in market shares and emission intensities captured by the within, between, and cross components to explain aggregate changes in the manufacturing sector emission intensity because they directly relate to our regression evidence for the effects of offshoring and Chinese import competition on sales and emission intensities presented in Section 4.2.³⁷ Therefore, row 2 decomposes the manufacturing sector emission intensity for the balanced panel such that the net entry component is zero. In the balanced sample, the overall decline in the manufacturing emission intensity is similar (55%) to the decline in the unbalanced panel (50%), and the within and between components still account about equally for the majority of this decline.

Existing research typically decomposes the overall emission intensity into industry-level components (Antweiler, Copeland, and Taylor 2001; Brunel 2017; Levinson 2009; Shapiro and Walker 2018). They find that adjustments within industries (called technique effects) account for most of the decline in aggregate emission intensities while adjustments across industries (composition effects) play a minor role. Interpreted through the lens of traditional comparative advantage trade theories such as the Heckscher-Ohlin model, the modest size of the between-industry changes seems to rule out trade as a potential explanation for declining emission intensities, since they predict a reallocation of economic activity across industries.

Alternatively, similar to us, a small set of papers perform decompositions at the firm level (Holladay and LaPlue III 2021; Najjar and Cherniwchan 2021). They refer to a process effect (within-firm changes in emission intensity), a reallocation effect (shifts in economic

³⁷It is difficult to consider the entry and exit components for two reasons. First, the biannual nature of the Industrial Energy Use Survey makes it hard to examine the entry of small firms. Second, our identification strategy requires a balanced sample, where all firms import before the shock as our instruments (defined in Section 3.3) work at the intensive margin of trade and would not be able to predict the entry and exit of firms.

activity across continuing firms), and a selection effect (firm entry and exit). Our estimates in Section 4.2 provide evidence that both firm-level market shares (sales) and emission intensities respond to comparative advantage driven offshoring and import competition measures. Therefore, standard globalization forces may lead to a reallocation of firm-level market shares and changes in emission intensities that affect aggregate emission intensities instead of between or within industry-level adjustments. Consequently, we examine the relevance of these firm-level globalization channels for the aggregate manufacturing emission intensity in the following subsection.

5.2 Counterfactual Decomposition Results

In this subsection we use our estimates of the effects of offshoring and Chinese import competition on firm-level emission intensities and market shares from Section 4.2 to calculate counterfactual declines in the overall emission intensity in the hypothetical situations where offshoring or Chinese import competition are held fixed at their initial 1996 level.³⁸ This partial equilibrium exercise quantifies the role of within and cross-firm globalization channels for aggregate emission intensities.

As our benchmark, row A of Table 11 reproduces the decomposition from Table 10 for the balanced sample. Rows B-E report what the total change in the aggregate emission intensity and its decomposed components would have been for four counterfactual scenarios: no change in the firm-level emission intensity due to offshoring (B), no change in firm-level sales due to offshoring (C), no change in the firm-level emission intensity due to import competition (D), and no change in firm-level sales due to import competition (E). Standard errors are computed based on the standard errors of our estimates in Table 3 applying the delta method.

Row B shows that the overall decline in emission intensity would have been only 20 percent if the impact of offshoring on firm-level emission intensities had been switched off.

³⁸Details about how we calculate the counterfactual decompositions are provided in Appendix B.1. Najjar and Cherniwchan (2021) and Ahmadi, Yamazaki, and Kabore (2022) apply a similar approach to quantify the aggregate implications of climate policy. Appendix Table C7 performs the decomposition for three shorter periods of roughly seven years each. The decompositions are in line with decompositions over the full 20-year period.

In other words, we consider offshoring as if it had not changed since 1996 and as if it affected only firm-level emission intensities—the within-firm component. It is evident from Table 11 that the within-firm component accounts for most of the change as expected.³⁹

Taking as given the parameter estimates from Table 3, the decomposition reveals what the changes would have been in our sample of firms. However, according to the delta method, the standard errors of our parameter estimates translate into relatively high standard errors for the components of the decomposition and they should be interpreted with caution.

Table 11: Counterfactual Decompositions

Scenario	Total	Within firm	Between firm	Cross
(A) Actual	-0.55	-0.28	-0.29	0.02
(B) Counterfactual emission intensity with offshoring at initial level	-0.20 (0.28)	0.08 (0.26)	-0.29 (-)	0.01 (0.15)
(C) Counterfactual sales with offshoring at initial level	-0.42 (0.03)	-0.28 (-)	-0.12 (0.08)	-0.02 (0.02)
(D) Counterfactual emission intensity with CIP at initial level	-0.54 (0.00)	-0.27 (0.00)	-0.29 (-)	0.02 (0.00)
(E) Counterfactual sales with CIP at initial level	-0.56 (0.00)	-0.28 (-)	-0.31 (0.01)	0.03 (0.00)

Notes: The table shows decompositions of the relative change in the aggregate manufacturing sector emission intensity as defined in equation (9). The decompositions rely on a balanced sample of firms operating in 1996 and 2016, so the net entry effect is zero by construction. The counterfactual scenarios rely on the methodology explained in appendix B.1. Counterfactual values are calculated with parameter estimates from Table 3. Standard errors in parentheses are calculated using the delta method.

Row C shows the implications of eliminating the impact of offshoring on firm-level sales. Offshoring increases sales and thus reshuffles market shares across firms depending on their firm-specific offshoring exposure, and, in turn, this impacts the between-firm component depending on initial firm-level emission intensities. The overall decline in the emission intensity would have been 42 percent (76 percent of its actual decline) if offshoring had been at its initial level throughout. That is, firms that grow due to offshoring tend to be relatively clean such that dirty firms are left with lower market shares, which in total leads to a lower overall manufacturing emission intensity.⁴⁰ As expected, the between-firm component is responsible

³⁹Without the contribution from changes in offshoring, the within-firm component would have increased. The reason is that, to generate the counterfactual observations, we simply subtracted the contribution of offshoring from each observation, and emission intensities change for other reasons than offshoring. That said, given the magnitude of the standard error, we interpret this with caution.

⁴⁰Firms that see their market shares fall in response to offshoring have on average an emission intensity of 0.043, while firms that gain market shares have on average an emission intensity of 0.008.

for a large part of the effect, but the cross term also adds a small contribution.

Taken together, these results suggest that offshoring has played a large role in reducing the domestic manufacturing emission intensity. Over the time period considered, 1996 to 2016, a substantial relocation of economic activity happened through intermediate inputs being sourced from abroad. At the same time, sales of final goods increased.⁴¹ That is, emissions per dollar of output have fallen and total output has increased, which is also reflected in the fact that the remaining domestic production relies more on cleaner energy inputs such as electricity as documented in Tables 4 and 5. In the case of Denmark these changes jointly explain a substantial part of the decline in the aggregate manufacturing emission intensity according to our quantification exercise. We should emphasize that this does not mean that environmental regulation has not played a role in the clean-up of manufacturing production. For example, Shapiro and Walker (2018) and Najjar and Cherniwchan (2021) show for larger economies with lower trade intensities than Denmark (the U.S. and Canada), that regulation has played an important role in reducing air pollution. In addition (as mentioned above), Colmer et al. (2025) shows for French firms that the introduction of the EU Emissions Trading Scheme, which also applied to Danish firms⁴², contributed to reducing manufacturing carbon emissions by 14-16%. Instead, we take our results as evidence that offshoring is more important than previously documented at least for a small open economy such as Denmark.

Row D reports the counterfactual when we consider emission intensities as if firm-level Chinese import competition remained at 1996 levels. Keeping Chinese import competition unchanged at its initial 1996 level only has a small impact on the within component compared to the within component in the actually observed data in row A, amounting to roughly one percentage point of the decline. Thus, the largest increases in Chinese import competition appear to mainly hit small firms, while larger firms are relatively unaffected.

Finally, Chinese import competition also affects emissions of firms through its impact on sales. The between-firm component in the decomposition could then potentially contribute

⁴¹See Figure D3 for aggregate changes in emissions, output, and offshoring in the Danish manufacturing sector.

⁴²Throughout our sample window, Denmark has had a low but constant carbon tax levied on non-ETS firms.

to changes in the overall manufacturing emission intensity through changes in Chinese import competition. The between-firm component changes in response to Chinese import competition only if firms are differentially exposed to import competition (thus implying a reshuffling of market shares) and if they differ in their initial emission intensities (see the definition of the between firm component in equation (9)). That is, Chinese import competition reduces the overall emission intensity through the between-firm component if firms in the dirtiest industries are hit harder by increased competition and vice versa.

Firms are indeed exposed differentially to Chinese import competition. Firms in the cleanest industries, such as textiles and electrical equipment (see Figure 1) experienced the largest increases in Chinese import competition, while relatively dirty industries, e.g., pharmaceuticals and non-metallic minerals, have seen smaller increases in exposure to Chinese import competition. Figure 4 suggests that there is a negative relationship between initial emission intensities and the change in Chinese import competition.⁴³ However, row E shows that if Chinese import competition is held unchanged at its initial level, the overall manufacturing emission intensity is largely unchanged compared to the change in the actual data reported in row A (-0.56 v. -0.55).⁴⁴ Thus, the relationship between exposure and initial emission intensities is not strong enough to substantially change the overall manufacturing emission intensity.⁴⁵

5.3 Carbon Leakage and Global Emissions

We now examine emission leakage rates due to import competition from China and offshoring. We measure import competition and offshoring at the firm level. Therefore, we provide summary statistics based on firm-specific leakage rates.

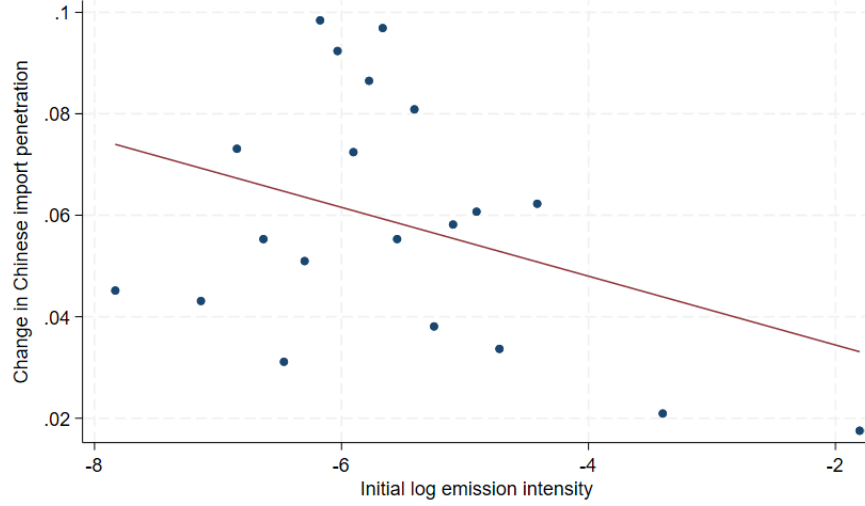
Let ΔE_i^{DK} be the change in firm i 's emissions in Denmark determined by its offshoring activities or import competition. Furthermore, let ΔE_i^{ROW} be the change in global emissions

⁴³Firms that see their market shares fall in response to Chinese import competition have, on average, an emission intensity of 0.008, while firms that gain market shares have, on average, an emission intensity of 0.057.

⁴⁴This result is robust to allowing for the effect heterogeneity of Chinese import competition found in Table 6, see Appendix Table C6.

⁴⁵Appendix Table C6 also shows that the results are robust to using the estimated coefficients from the separate regressions in Table 3 instead of the main results in Table 3 where the impact of the globalization variables is estimated jointly.

Figure 4: Changes in Chinese import penetration and initial emission intensities



Notes: The figure shows a binned scatter plot of long differences in firm-level Chinese import penetration against firms' initial (log) emission intensity. The regression line is based on the observations in the balanced panel of Table 10 (the slope coefficient is -0.0068 with a t-value of -1.83.)

contained in i 's offshore activities or import competition. Then, we define the carbon leakage rate as the tons of carbon created abroad relative to the change in emissions at home:

$$L_i = -\frac{\Delta E_i^{ROW}}{\Delta E_i^{DK}}. \quad (10)$$

Appendix B.2 explains in detail how we calculate the leakage rates for offshoring and Chinese import competition. The intuition is straightforward. Starting with offshoring, the numerator is approximated by calculating the carbon content embodied in each firm's offshoring activity. To do so, we follow Shapiro (2021) and Copeland, Shapiro, and Taylor (2022) and measure the emissions generated abroad contained in each firm's import flows using emission intensities by broad product categories and origin countries from Exiobase.⁴⁶ Note that the origin countries of Danish firms' offshoring activities are mainly European. For example, the top three origin countries, Germany, Sweden and Norway, account for more than 40% of all intermediate input import flows, see Appendix Table A4. We approximate

⁴⁶Exiobase is an environmentally extended multi-regional input-output database from which emission coefficients capturing both direct emissions and indirect emissions from supply chains are derived. To be consistent, we not only calculate emissions embodied in imports using Exiobase emission coefficients, but we also use them to obtain a measure for local emissions in the denominators of the leakage rates). We provide more details about Exiobase and how we implement it in our context in Appendix A.

the domestic reductions in carbon emissions due to offshoring, the denominator, based on the estimated offshoring elasticity of emissions in column (7) of Table 3, and firms' change in offshoring activities. Similarly, to generate leakage rates associated with import competition, we compute changes in emissions in China based on import flows and Exiobase emission coefficients. We approximate the domestic reductions in carbon emissions due to import competition, the denominator, using the estimated import-competition elasticity of emissions in column (7) of Table 3, and firm-level changes in import competition.⁴⁷

We find that the offshoring leakage rate for the median firm is 4. That is, when offshoring reduces emissions in the median firm by 1 ton in Denmark, emissions increase by roughly four tons abroad. However, this number is imprecisely estimated due to the high standard error of the offshoring elasticity of column (7) in Table 3.⁴⁸ That is, we cannot rule out that emissions generated abroad due to offshoring have the same order of magnitude as the emission reduction at home.

By contrast, the leakage rate for Chinese import competition is considerably higher at 144 for the median firm. This means that when Chinese import competition reduces emissions in Denmark through lower sales and lower emission intensities, emissions in China increase by a substantially larger amount. This result is partly driven by the fact that Exiobase emission intensities in China are much higher than the corresponding intensities in Denmark, see Figure 5.

Another reason is that import competition from China only has a rather modest displacement effect on Danish emissions, which tends to generate a low value of the denominator in the leakage rate as defined in equation (10).⁴⁹ In other words, this tends to produce a high leakage rate.

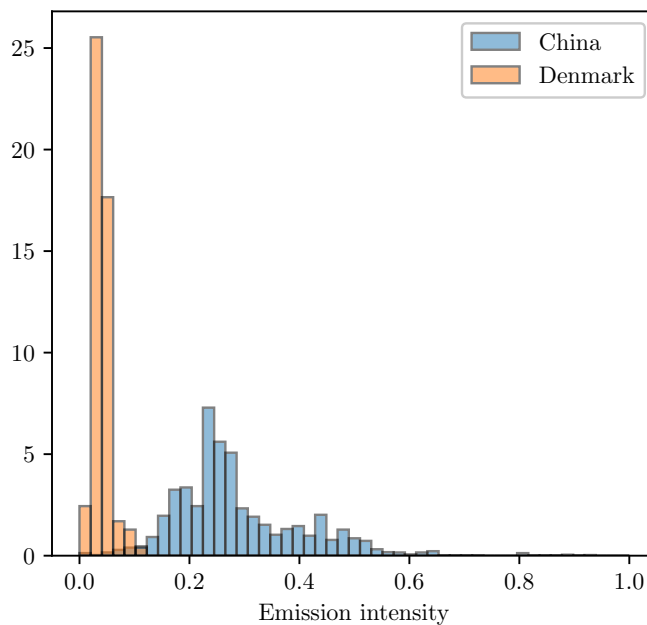
For context, the literature estimating leakage rates for carbon taxes typically find leakage rates of at most 10 (Copeland, Shapiro, and Taylor 2022). There are several potential rea-

⁴⁷The leakage rates do not account for potential cross effects, that is, emission intensities potentially changing due to offshoring at the same time the scale of production changes due to import competition. The decomposition in Tables 10 and 11 provide little evidence for cross effects and estimating interaction effects in our empirical specification is challenging because it requires three instruments for three endogenous variables.

⁴⁸Appendix Table C8 shows the distribution of leakage rates across firms and their standard errors.

⁴⁹Evaluated at the median value of the Chinese import penetration variable from Table 1, the semi-elasticity for Chinese import competition from column (7) in Table 3 corresponds to an elasticity of -0.034.

Figure 5: Histogram of emission intensities in Denmark and China



Notes: The figure compares emission intensities between China and Denmark at the Exiobase product level. The emission intensity of a given Exiobase product is the across-year average. The included HS6 products are those that enter at least one firm's import penetration measure. Some HS6 products are not represented in Exiobase and are excluded. The histogram is normalized so it integrates to one.

sons why our leakage rates based on import competition are higher. Carbon taxes directly target dirty production, and, consequently, likely displace domestic emissions. By contrast, the globalization effects we estimate are driven by changes in other countries' comparative advantage that are likely less correlated with emission intensities in domestic (Danish) production. This tends to produce lower values in the denominator of the leakage rates. Finally, import competition focuses on China, one of the most emission-intensive countries in the world. Consequently, increasing imports associated with import competition from China generates substantial changes in emissions abroad, the numerator of the leakage rate.

6 Conclusion

As noted by Copeland, Shapiro, and Taylor (2022) relatively little work has provided evidence for the causal impact of globalization on environmental outcomes, and this paper contributes to filling this gap. We use Danish firm-level data for carbon emissions and globalization exposure as measured by offshoring and Chinese import competition. To set the stage, we

first decompose the decline in the manufacturing sector emission intensity into within-firm, between-firm and cross components. We then estimate the causal impacts of offshoring and Chinese import competition on firm-level emission intensities and sales. We use the estimation results to quantify the contribution of offshoring and Chinese import competition to the overall manufacturing sector emission intensity. Finally, we also use the estimation results to calculate carbon leakage rates to assess the consequences for global emissions.

To summarize our basic regression results, we find that, one, offshoring has counteracting effects on firm-level emissions. Firms become cleaner per unit of output, but they also become more productive, larger and emit more as a result. We are the first to highlight these counteracting effects, and the net effect of offshoring on the manufacturing sector emission intensity is unclear without further investigation. Two, we are the first to jointly examine how Chinese import competition affects the overall domestic manufacturing sector emission intensity through changes in within- and between-firm components. We find that import competition affects emissions through changes in emission intensities within firms over time as well as reallocation of output across firms. Again, the net effect on the overall domestic manufacturing sector emission intensity is unclear without further scrutiny as it depends on initial firm-level emission intensities and firm-level exposure to import competition. Three, we examine offshoring and Chinese import competition in a coherent framework and show that these two driving forces behind the growth in international trade have clear (but quite distinct) implications for firm-level carbon emission intensities and sales. This is potentially important for how the overall manufacturing sector emission intensity changes in response to international trade.

We then compute how firm-level emission intensities and sales would have evolved had they not been affected by offshoring and Chinese import penetration and relate it to our decomposition exercise. We show that offshoring has contributed substantially to the overall decline in the manufacturing emission intensity through both the within- and between-firm components. On the other hand, Chinese import competition does not affect the overall manufacturing sector emission intensity because firm-level exposure to import competition is only weakly correlated with initial emission intensities. Thus offshoring and Chinese import competition have very different implications for the domestic manufacturing sector

emission intensity.

Finally, we show that offshoring and Chinese import competition also have very different implications for global carbon emissions. We calculate embodied carbon emissions in offshoring and import flows and use them to derive carbon leakage rates based on our regression results. We cannot rule out that emissions embodied in imports of intermediate inputs are of roughly the same in magnitude as the domestic emission reduction caused by the intermediate input import flows. By contrast, emissions embodied in final good imports from China are much larger in magnitude than the domestic emission reduction. Thus, overall, import competition from China strongly increases global carbon emissions.

References

- Aghion, Philippe, Antonin Bergeaud, Matthieu Lequien, Marc J. Melitz, and Thomas Zuber. 2024. Opposing firm-level responses to the China shock: Output competition versus input supply. *American Economic Journal: Economic Policy* 16:249–269.
- Ahmadi, Younes, Akio Yamazaki, and Philippe Kabore. 2022. How do carbon taxes affect emissions? Plant-level evidence from manufacturing. *Environmental and Resource Economics* 82:285–325.
- Akerman, Anders, Rikard Forslid, and Ossian Prane. 2024. Imports and the CO2 Emissions of Firms. *Journal of International Economics* 152 (1): 104004.
- Amiti, Mary, and Jozef Konings. 2007. Trade liberalization, intermediate inputs, and productivity: Evidence from Indonesia. *American Economic Review* 97 (5): 1611–1638. <https://doi.org/10.1257/aer.97.5.1611>.
- Antweiler, Werner, Brian R. Copeland, and M. Scott Taylor. 2001. Is free trade good for the environment? *American Economic Review* 91:877–908.
- Autor, David, David Dorn, Gordon H. Hanson, Gary Pisano, and Pian Shu. 2020. Foreign competition and domestic innovation: Evidence from US patents. *American Economic Review: Insights* 2 (3): 357–374. <https://doi.org/10.1257/aeri.20180458>.
- Autor, David H., David Dorn, and Gordon H. Hanson. 2013. The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review* 103 (6): 2121–68.
- . 2016. The China shock: learning from labor-market adjustment to large changes in trade. *Annual Review of Economics* 8:205–240. <https://doi.org/10.1146/annurev-economics-080315-015041>.
- Bartik, Timothy J. 1991. Who benefits from state and local economic development policies? Kalamazoo, MI: WE Upjohn Institute for Employment Research.

- Bernard, Andrew B., Teresa C. Fort, Valerie Smeets, and Frederic Warzynski. 2025. Heterogeneous globalization: Offshoring and reorganization. Working Paper 26854. National Bureau of Economic Research.
- Bernard, Andrew B., J. Bradford Jensen, Stephen J. Redding, and Peter K. Schott. 2018. Global firms. *Journal of Economic Literature* 56:565–619.
- Bernard, Andrew B., J. Bradford Jensen, and Peter K. Schott. 2006. Survival of the best fit: Exposure to low-wage countries and the (uneven) growth of US manufacturing plants. *Journal of International Economics* 68:219–237.
- Bloom, Nicholas, Mirko Draca, and John Van Reenen. 2015. Trade induced technical change? The impact of Chinese imports on innovation, IT and productivity. *The Review of Economic Studies* 83 (1): 87–117.
- . 2016. Trade induced technical change? The impact of Chinese imports on innovation, IT and productivity. *Review of Economic Studies* 83 (1): 87–117. <https://doi.org/10.1093/restud/rdv039>.
- Bombardini, Matilde, and Bingjing Li. 2020. Trade, pollution and mortality in China. *Journal of International Economics* 125:103321.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2021. Quasi-experimental shift-share research designs. *The Review of Economic Studies* 89:181–213.
- Brunel, Claire. 2017. Pollution offshoring and emission reductions in EU and US manufacturing. *Environmental and Resource Economics* 68:621–641.
- Cherniwchan, Jevan. 2017. Trade liberalization and the environment: Evidence from NAFTA and US manufacturing. *Journal of International Economics* 105:130–149.
- Cherniwchan, Jevan, Brian R. Copeland, and M. Scott Taylor. 2017. Trade and the environment: New methods, measurements, and results. *Annual Review of Economics* 9:59–85.
- Choi, Jaerim, Jay Hyun, Gueyon Kim, and Ziho Park. 2025. The cleanup of US manufacturing through pollution offshoring. *Journal of International Economics* 154:104046.

- Cole, Matthew A., Robert J. R. Elliott, and Toshihiro Okubo. 2014. International environmental outsourcing. *Review of World Economics* 150:639–664.
- Cole, Matthew A., Robert J. R. Elliott, Toshihiro Okubo, and Liyun Zhang. 2021. Importing, outsourcing and pollution offshoring. *Energy Economics* 103.
- Colmer, Jonathan, Ralf Martin, Mirabelle Muûls, and Ulrich J. Wagner. 2025. Does pricing carbon mitigate climate change? Firm-level evidence from the European Union emissions trading scheme. *Review of Economic Studies* 92:1625–1660.
- Copeland, Brian R., Joseph S. Shapiro, and M. Scott Taylor. 2022. “Globalization and the environment.” In *Handbook of International Economics, Volume 5*, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff. Elsevier.
- Dauth, Wolfgang, Sebastian Findeisen, and Jens Suedekum. 2014. The rise of the east and the far east. *Journal of the European Economic Association* 12:1643–1675.
- Dussaux, Damien, Francesco Vona, and Antoine Dechezleprêtre. 2023. Imported carbon emissions: Evidence from French manufacturing companies. *Canadian Journal of Economics* 56 (2): 593–621.
- Forslid, Rikard, Toshihiro Okubo, and Karen Helene Ullveit-Moe. 2018. Why are firms that export cleaner? International trade, abatement and environmental emissions. *Journal of Environmental Economics and Management* 91:166–183.
- Foster, Lucia, John C. Haltiwanger, and C. J. Krizan. 2001. “Aggregate Productivity Growth: Lessons from Microeconomic Evidence.” In *New Developments in Productivity Analysis*, edited by Charles R. Hulten, Edwin R. Dean, and Michael J. Harper, 63:303–372. Studies in Income and Wealth. University of Chicago Press.
- Grossman, Gene M., and Esteban Rossi-Hansberg. 2008. Trading tasks: A simple theory of offshoring. *American Economic Review* 98, no. 5 (December): 1978–97. <https://doi.org/10.1257/aer.98.5.1978>. <https://www.aeaweb.org/articles?id=10.1257/aer.98.5.1978>.

- Gutiérrez, Emilio, and Kensuke Teshima. 2018. Abatement expenditures, technology choice, and environmental performance: Evidence from firm responses to import competition in Mexico. *Journal of Development Economics* 133:264–274.
- Holladay, J. Scott, and Lawrence D. LaPlue III. 2021. Decomposing changes in establishment-level emissions with entry and exit. *Canadian Journal of Economics* 54:1046–1071.
- Hummels, David, Rasmus Jørgensen, Jakob R. Munch, and Chong Xiang. 2014. The wage effects of offshoring: Evidence from Danish matched worker-firm data. *American Economic Review* 104, no. 6 (June): 1597–1629. <https://doi.org/10.1257/aer.104.6.1597>. <http://www.aeaweb.org/articles?id=10.1257/aer.104.6.1597>.
- Hummels, David, Jakob R. Munch, and Chong Xiang. 2018. Offshoring and labor markets. *Journal of Economic Literature* 56 (3): 981–1028. <https://doi.org/10.1257/jel.20161150>. <https://www.aeaweb.org/articles?id=10.1257/jel.20161150>.
- Johnson, Robert C., and Guillermo Noguera. 2012. Accounting for intermediates: Production sharing and trade in value added. *Journal of International Economics* 86:224–236.
- Levinson, Arik. 2009. Technology, international trade, and pollution from US manufacturing. *American Economic Review* 99 (5): 2177–92.
- Lyubich, Eva, Joseph Shapiro, and Reed Walker. 2018. Regulating mismeasured pollution: Implications of firm heterogeneity for environmental policy. *American Economic Review, Papers and Proceedings* 108:136–42.
- Marin, Giovanni, and Francesco Vona. 2021. The impact of energy prices on socioeconomic and environmental performance: Evidence from French manufacturing establishments, 1997–2015. *European Economic Review* 135:103739.
- Martin, Leslie A. 2011. Energy efficiency gains from trade: Greenhouse gas emissions and India’s manufacturing sector. Working paper. UC Berkeley.
- Melitz, Marc J. 2003. The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica* 71 (6): 1695–1725.

- Melitz, Marc J., and Sašo Polanec. 2015. Dynamic Olley-Pakes productivity decomposition with entry and exit. *The Rand Journal of Economics* 46 (2): 362–375.
- Metcalf, Gilbert E. 2019. On the economics of a carbon tax for the United States. *Brookings Papers on Economic Activity* Spring:405–484.
- Mion, Giordano, and Linke Zhu. 2013. Import competition from and offshoring to China: A curse or blessing for firms? *Journal of International Economics* 89:202–215.
- Najjar, Nouri, and Jevan Cherniwchan. 2021. Environmental regulations and the cleanup of manufacturing: Plant-level evidence. *The Review of Economics and Statistics* 103, no. 3 (July): 476–491. https://doi.org/10.1162/rest_a_00904. <https://ideas.repec.org/a/tpr/restat/v103y2021i3p476-491.html>.
- Pavcnik, Nina. 2002. Trade liberalization, exit, and productivity improvements: Evidence from Chilean plants. *Review of Economic Studies* 69 (1): 245–276. <https://doi.org/10.1111/1467-937X.00201>.
- Rodrigue, Joel, Dan Sheng, and Yong Tan. 2024b. Exporting, abatement, and firm-level emissions: Evidence from China’s accession to the WTO. *Review of Economics and Statistics* 106 (4): 1064–1082.
- Shapiro, Joseph S. 2021. The environmental bias of trade policy. *The Quarterly Journal of Economics* 136, no. 2 (December): 831–886. <https://doi.org/10.1093/qje/qjaa042>. <https://doi.org/10.1093/qje/qjaa042>.
- Shapiro, Joseph S., and Reed Walker. 2018. Why is pollution from US manufacturing declining? The roles of environmental regulation, productivity, and trade. *American Economic Review* 108, no. 12 (December): 3814–54. <https://doi.org/10.1257/aer.20151272>. <http://www.aeaweb.org/articles?id=10.1257/aer.20151272>.
- Stadler, Konstantin, Richard Wood, Tatyana Bulavskaya, Carl-Johan Södersten, Moana Simas, Sarah Schmidt, Arkaitz Usubiaga, José Acosta-Fernández, Jeroen Kuenen, Martin Bruckner, et al. 2018. EXIOBASE 3: Developing a time series of detailed environ-

mentally extended multi-regional input-output tables. *Journal of Industrial Ecology* 22 (3): 502–515.

Wagner, Ulrich J., Dalia Kassem, Andreas Gerster, Justina Jaraite-Kazukauske, Marit E. Klemetsen, Marita Laukkanen, Johannes Leisner, et al. 2020. Carbon footprints of european manufacturing jobs: Stylized facts and implications for climate policy. Discussion Paper No. 250. Epos Collaborative Research Center.

Appendices

A Data Appendix

A.1 Coverage of Regression Sample

For the regression sample, we require all firm-years to have positive emissions, more than 300.000 DKK offshoring, positive sales, positive world export supply (*WES*) instrument and defined Chinese export supply (*CES*) instrument. The resulting number of firm-years is 11,591 across 2019 unique firms. Table A1 shows that the sample covers most of total carbon emissions (72.6 percent) and economic activity (roughly two thirds of sales and value added).

Table A1: Share of Economic Activity in Sample

Annual Emissions	% of Value-added	% of Employment	% of Sales	% of Imports	% of CO2
7,525,363.64	63.38	59.28	66.56	77.16	72.55

Notes: This table considers firms in manufacturing and compares firm characteristics for the firms in the regression sample with the entire population of manufacturing firms. The table only includes firms with 20 or more employees (except for aggregate emissions, see below). The table reports how large a share of e.g. employment sample firms account for. The total emissions reported here are the aggregate manufacturing emissions including biomass as reported by Statistics Denmark. Each cell reports an across-year average.

A.2 Firm-Level Carbon Emissions

To calculate firm-level carbon emissions, we measure for each firm in the regression sample the energy use of various fuels from the Industrial Energy Use Survey, multiply each fuel by its physically determined emission coefficient, and sum over fuels. We also add 'process emissions', that is, emissions related to the production process rather than the burning of fuels.

The fuel-specific emission coefficients rely on data from Statistics Denmark. Each emission coefficient measures CO2 emissions per giga joule of energy generated from burning the fuel. Coefficients are constant across sectors and time because physically, the calorific value of the fuel as well as the emissions per unit of mass is constant.⁵⁰ Metcalf (2019) puts it

⁵⁰One exception in our data is the fuel 'general waste', since its components shift over time. Still, the time variation for this fuel is very limited.

this way: "The amount of CO₂ associated with burning a ton of coal, a gallon of gasoline, or a therm of natural gas is, for all intents and purposes, constant. Changes in industrial processes may affect the amount of fossil fuel burned but not the emissions per unit of fuel input."

The process emissions are measured at an aggregated sector level (a Danish sector definition slightly more granular than the 2-digit NACE industries). We distribute these process emissions onto firms in the relevant industry in proportion to the individual firm's share of the total sales in that industry in that year. Appendix Table C5 shows that our main results are robust to leaving out process emissions from the firm level carbon emission measure.

Table A2 shows the average share that the emissions from individual fuels and process emissions constitute of the average firm's emissions. For example, it shows that natural gas is the most important energy input, and that process emissions amount to 10 percent of the average firm's emissions.

Table A2: Average emission share of inputs and process emissions

Type	Average share
Natural gas	0.45
Gas diesel	0.18
Gasoline	0.13
Process emissions	0.10
Liquified petroleum gas	0.04
Fuel oil	0.04
Remainder	0.05

Notes: The table shows the average emission share of various energy inputs and process emission across all firm-years in the sample. The type called 'Remainder' refers to all remaining energy inputs: Biodegradable waste, wood pellets, additional gas types such as biogas and refinery gas, general waste, coal, petroleum, coke and additional oil types.

A.3 Firm-Level Importing Statistics

Table A3 presents summary statistics of the importing and offshoring behavior of the firms in the regression sample. The table takes all firm-year-origin-product import flows, where origin refers to the source country and product refers to an HS6 code, and calculates various

summary statistics.

Table A3: Summary statistics for firm-level imports

	All import flows	Offshoring flows only
Panel A: Totals		
Total unique products (HS6-codes)	5530	5107
Avg. yearly unique origin-products	22,865	19,674
Total unique origin-products	72,998	63,136
Panel B: Origin-product-level		
Number of firms importing an origin-product, median product	1.00	1.00
Number of firms importing an origin-product, average product	2.59	2.23
Panel C: Firm-level number of products		
Number of origin-products, median firm	34.39	22.06
Number of origin-products, average firm	58.42	43.32
Panel D: Share of total value of flows		
2 most imported origin-products, median firm	0.54	0.61
5 most imported origin-products, median firm	0.80	0.86
Pre-sample origin-products, aggregate	0.54	0.56
Panel E: Share of imports		
Offshoring, aggregate	0.92	-
Offshoring, median firm	0.95	-

Notes: This table presents calculations from firm-year-origin-product import flows. An origin-product is a combination of an origin country and an HS6 product code. All panels except Panel A (rows 1 and 3) calculate the stated statistic for each year separately and then reports the across-year average. The columns indicate the set of import flows (all or only those categorized as offshoring) used to calculate the statistic.

Panel B calculates for each origin-product, how many firms that import it, and presents the median and average products.

Panel C calculates the number of unique origin-products that a firm has in a given year, and then reports the median and average firm respectively.

Panel D, the first two rows, take each firm-year and calculates the share that the 2/5 most imported origin-products have out of that firm-year's total imports and offshoring respectively, and reports those statistics for the median firm. The third row of panel D reports the fraction of total values that the flows of pre-sample origin-products account for. The pre-sample origin-products are firm-specific and correspond to the origin-products that a firm has imported in the two years prior to the first year that the firm enters the regression sample. These pre-sample origin-products correspond to those used for the shares in our Bartik shift-share instrument for our IV-regressions.

Panel E calculates the fraction of total imports (either at the aggregate or inside each firm-year) that offshoring flows constitute.

Panel A calculates the total number of HS6 products as well as the total and across-year average of origin-product combinations. This shows that the 2019 firms observed during the sample window import 22,865 unique origin-products in an average year. When considering offshoring flows only, this number drops to 19,674 and is still large relative to the number

of firms. That the number of origin-products is large relative to the number of firms is also reflected in panel B: The average origin-product is imported by only 2.59 firms or 2.23 firms when considering offshoring flows. For the median product, the number is 1 for both imports and offshoring. That is, the importing behavior of Danish firms is highly dispersed across different source countries and products. This is useful for our identification strategy as it provides variation across firms due to a supply shock to a particular origin-product that only affects few Danish firms directly.

Panel C shows the number of origin-products in all firms for the median and average firm. The median firm imports 34 origin-products for all imports but only 22 origin-products for offshoring flows. This limits the extent of overlap between the origin-products that firms import. Panel D calculates how large a share of import or offshoring flows that the 2 or 5 most imported products cover for each firm individually. For example, for the median firm, the 5 most important origin-products cover 86 percent of total offshoring. Essentially, the imports of a particular firm are concentrated in just a few origin-products.

A second important calculation from panel D is its third row. It identifies the origin-products that a firm imported during its pre-sample years, i.e., the two years preceding the first year that a given firm has emission data. Then, it reports how large a share of the total import/offshoring value that these origin-products cover for the flows in the regression sample. Importantly for our identification strategy, the share is large for offshoring (0.56), reflecting a relatively stable importing behavior of firms. This stability is an important reason why the instrument is a strong predictor of offshoring.

Finally, panel E reports that offshoring flows cover 92 percent of the aggregate import flows and 95 percent of the firm level imports for the median firm. This reflects that firms' imports are in fact mainly in product categories that they themselves produce. However, there is substantial variation across firms. For example, the 25th percentile is 82 percent.

In summary, Table 1 and Table A3 document significant within-firm variation in offshoring and also that the few products that make up offshoring are highly specific to individual firms.

Table A4 shows the source country distribution for offshoring in the sample. The top ten source countries account for more than 70 percent of intermediate input import flows, and

it is evident that proximate countries dominate with Germany as a particularly important source country for Danish firms. It is also seen that China accounts for a modest 3 percent of total offshoring flows.

Table A4: Top 10 offshoring origin countries

Origin country	Share of Offshoring
Germany	21.36
Norway	12.38
Sweden	9.34
United kingdom	5.89
Netherlands	5.22
USA	3.94
Belgium	3.29
France	3.22
Italy	3.10
China	2.95

Notes: The table shows the percentage of total offshoring that is accounted for by specific origin countries. The reported share is the across year average among all sample years.

Finally, Table A5 shows additional summary statistics for how our firm-level Chinese import competition measure varies across industries and across firms within industries. Domestic firms in industries such as Textiles, Wearing Apparel, Leather and Furniture are on average exposed more to Chinese import competition firms in other industries. It is also evident that exposure varies a lot across firms within industries depending on their initial product mix. For example, furniture producers at the 75th percentile are roughly six times more exposed to import competition from China than producers at the 25th percentile (a weighted Chinese import market share of 23.5% vs. 3.7%).

Table A5: Chinese Import Competition Across and Within Industries

2-digit NACE industry	Avg.	Std.	p25	p50	p75	N
Food Products	0.006	0.018	0	0	0.004	1345
Beverages	0.001	0.001	0	0.001	0.002	84
Tobacco	0	0.001	0	0	0	46
Textiles	0.121	0.147	0.004	0.056	0.193	321
Wearing Apparel	0.152	0.110	0.073	0.120	0.213	238
Leather	0.100	0.143	0.006	0.027	0.142	56
Wood	0.016	0.030	0	0.001	0.021	485
Paper	0.022	0.035	0.001	0.008	0.029	339
Media Printing						12
Coke and Refined Petroleum						34
Chemicals	0.016	0.045	0	0.001	0.009	483
Pharmaceuticals	0.009	0.021	0	0.001	0.004	154
Rubber and Plastic	0.049	0.060	0.004	0.034	0.072	1078
Non-Metallic Mineral	0.047	0.115	0	0.001	0.028	508
Basic Metals	0.039	0.069	0	0.004	0.047	232
Metal Products	0.065	0.086	0.003	0.032	0.091	1351
Electronic Products	0.064	0.086	0.005	0.023	0.090	759
Electrical Equipment	0.056	0.079	0.004	0.024	0.074	422
Machinery	0.033	0.061	0.001	0.009	0.037	2364
Motor Vehicles	0.028	0.065	0	0.004	0.023	332
Other Transportation	0.031	0.080	0	0.005	0.021	96
Furniture	0.140	0.116	0.037	0.105	0.235	511
Other	0.083	0.124	0.003	0.023	0.117	341

Notes: The table shows values of the Chinese import competition measure, CIP_{it} , defined in equation (7). The summary statistics are based on all firm-year observations in the sample.

A.4 Exiobase

Calculating carbon leakage rates for offshoring and Chinese import competition requires internationally comparable emission intensities. To measure such emission intensities, we use the environmentally extended multi-regional input-output (EE MRIO) tables of the Exiobase 3 database (Stadler et al. 2018). Exiobase uses national accounts, trade statistics, national input-output tables and emission accounts from The International Energy Agency as its sources, among others. International consistency was prioritized in the construction of the database. This is important for our analysis, since a leakage rate is inherently a cross-country metric.

Exiobase contains 44 countries, including all EU countries and 16 additional major economies including China, plus an additional 5 "rest of world" regions. The database covers roughly 90 percent of global GDP. It contains 200 product categories across all sectors of the economy and measures variables such as output, emissions and the corresponding emission intensities for all years from 1995 to 2016. The emission intensities measure CO₂ emissions from fossil fuel combustion as well as non-combustion emissions from cement and lime production per million euros of output for each country-product-year combination. We convert these into emissions per DKK from emission per million Euros by using yearly averages of the exchange rate from EUR to DKK (extrapolating 1999, the initial year of the Euro, backwards to 1995) and deflating to 2015 as with other monetary variables. The structure of Exiobase allows us to use emission intensities which reflect indirect emissions embedded in the input-output linkages of the world economy. We map each of these emission intensities to each HS6-country-year in the foreign trade statistics and PRODCOM. When the mapping is not unique, we calculate a weighted average across Exiobase emission intensities using output in the particular product-country-years as weights. For a comparison of Exiobase to other EE MRIOs see the online appendix of Shapiro (2021).

A.5 Danish Expenditure Shares on Chinese Goods

For the years 2000, 2007, and 2016 we obtain Total Turnover (DKK 1,000) and Industrial Services (DKK 1,000) in the Danish manufacturing sector from Statistics Denmark.⁵¹ We compute turnover in manufacturing goods in DKK as total turnover net of industrial services.

We transform turnover in manufacturing on goods into US\$ using annual exchange rates reported by the IMF.⁵²

We compute total Danish expenditure on manufacturing goods as total turnover in Danish manufacturing goods plus total imports of manufacturing goods minus total exports of manufacturing goods. We obtain total manufacturing imports and exports from WITS.⁵³

Following Autor, Dorn, and Hanson (2013) we then compute China's import penetration in Denmark as manufacturing goods imports from China divided by the total Danish expenditure on manufacturing goods. Again, the source of the import data is WITS.

⁵¹<https://www.statbank.dk/statbank5a/selectvarval/define.asp?PLanguage=1&subword=tabel&MainTable=OMS5&PXSIId=178069&tablestyle=&ST=SD&buttons=0>

⁵²[https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:ER\(4.0.1\)](https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.STA:ER(4.0.1))

⁵³<https://wits.worldbank.org/WITS/WITS/Restricted/Login.aspx>

B Counterfactual Decomposition and Carbon Leakage

B.1 Counterfactual Decompositions

We wish to perform the decomposition of equation (9) in a way that allows us to assess how the overall manufacturing emission intensity would have evolved had offshoring or import competition stayed constant at their initial levels. To do so, we calculate counterfactual data series. For offshoring, we replace actual values of firm-level emission intensities by those predicted by the regression results of Table 3:

$$\log \tilde{\phi}_{i2}^{OFF} = \log \phi_{i2} - \hat{\beta} \Delta \log \text{OFF}_i, \quad (11)$$

where $\tilde{\phi}_{i2}^{OFF}$ denotes the counterfactual emission intensity in period 2, ϕ_{i2} is the actual emission intensity in period 2, and $\Delta \log \text{OFF}_i = \log \text{OFF}_{i2} - \log \text{OFF}_{i1}$ is the observed change in (log) offshoring for firm i . When predicting counterfactual sales for offshoring, we perform the corresponding calculation using the parameter estimate $\hat{\theta}$. That is

$$\log \tilde{S}_{i2}^{OFF} = \log S_{i2} - \hat{\theta} \Delta \log \text{OFF}_i, \quad (12)$$

where $\tilde{S}_{i2}^{OFF}$ and S_{i2} are counterfactual and observed sales respectively. After calculating counterfactual log sales, we can calculate the counterfactual market shares $\tilde{s}_{i2}^{OFF}$ in period 2 used in the decomposition simply by taking the exponential and dividing by the sum of (counterfactual) sales, i.e., $\tilde{s}_{i2}^{OFF} = \frac{\tilde{S}_{i2}^{OFF}}{\sum_j \tilde{S}_{j2}^{OFF}}$.

Similarly for import competition, we replace actual values of emission intensities and log sales by their predicted values when Chinese import competition is kept constant:

$$\log \tilde{\phi}_{i2}^{CIP} = \log \phi_{i2} - \hat{\lambda} \Delta \text{CIP}_i \quad (13)$$

$$\log \tilde{S}_{i2}^{CIP} = \log S_{i2} - \hat{\delta} \Delta \text{CIP}_i, \quad (14)$$

where $\Delta \text{CIP}_i = \text{CIP}_{i2} - \text{CIP}_{i1}$ is the observed change in firm-level Chinese import competition.

B.2 Carbon Leakage

Our estimation results show that offshoring reduces firm-level emission intensities but increases sales, and the counterfactual decomposition exercise showed that a substantial portion of the decline in the overall manufacturing emission intensity can be attributed to offshoring. Likewise, Chinese import competition reduces firm-level sales and emissions. Both of these phenomena, however, are associated with increasing emissions abroad. This section outlines how we quantify this relationship by calculating carbon leakage rates.

For any specific firm i we are interested in the change in global emissions coming from the reduction in Danish emissions (triggered by offshoring or Chinese import competition) and the associated increase in emissions in the rest of the world:

$$\Delta E_t^{\text{Global}} = \Delta E_t^{DK} + \Delta E_t^{ROW}. \quad (15)$$

We express this as a carbon leakage rate that measures the change in emissions in the rest of the world per unit of domestic emission reduction $\left(-\frac{\Delta E_t^{ROW}}{\Delta E_t^{DK}}\right)$. To calculate the terms underlying this fraction, we rely on Exiobase emission coefficients (see section A.4).

B.2.1 Foreign emissions

Emissions in the rest of the world associated with imports by Danish firms or industries are measured as the change in the value of imports at the country-product-time level multiplied by the corresponding emission coefficient:

$$\Delta E_{it}^{ROW} = \sum_c \sum_p \Delta I_{cpt} e_{cpt}, \quad (16)$$

where e_{cpt} refers to the emission intensity of product p from Exiobase. Since the products in Exiobase are more aggregated than the product level in imports (HS6), we aggregate import flows from HS6 into Exiobase-products before doing the calculation.

B.2.2 Offshoring

This subsection derives the leakage rate for offshoring, $L_i^O = -\Delta E_i^O / \Delta E_i$, where ΔE_i^O is the change in emissions embodied in firm i 's offshoring corresponding to ΔE_i^{ROW} above, and ΔE_i is the change in emissions from domestic firm i corresponding to ΔE_i^{DK} above (we suppress subscript t). To be consistent we not only calculate emissions embodied in imports using Exiobase emission coefficients, but we also use them to obtain a measure for local emissions in the denominators, i.e., the domestic emissions of the leakage rate.

To calculate the leakage rate for offshoring we first consider the numerator, that is, the change in foreign emissions induced by offshoring. The emissions embodied in offshoring are defined as $E_i^O = e_i^O OFF_i$, where e_i^O is the emission intensity in firm i 's offshoring, which is measurable from Exiobase as outlined above. The total differential, keeping the foreign emission intensity fixed, is $\Delta E_i^O = e_i^O \Delta OFF_i = \frac{E_i^O}{OFF_i} \Delta OFF_i$.

The denominator of the leakage rate is derived from totally differentiating domestic emissions of firm i , $E_i = e_i^Y Y_i$, where e_i^Y is the emission intensity in firm i 's output, Y_i . We then have $\Delta E_i = e_i^Y \Delta Y_i + Y_i \Delta e_i^Y$, where the first term comes from the productivity effect of offshoring and the second term from offshoring's impact on the firm-level emission intensity. We can then use the estimated output and emission intensity elasticities of offshoring to get

$$\Delta E_i = e_i^Y Y_i \underbrace{\frac{\Delta OFF_i}{OFF_i} \hat{\theta}}_{\frac{\Delta Y_i}{Y_i}} + Y_i e_i^Y \underbrace{\frac{\Delta OFF_i}{OFF_i} \hat{\beta}}_{\frac{\Delta e_i^Y}{e_i^Y}} = E_i \frac{\Delta OFF_i}{OFF_i} (\hat{\beta} + \hat{\theta}). \quad (17)$$

We can now plug into the leakage rate and, writing year subscripts explicitly, obtain

$$L_{it}^O = -\frac{\Delta E_{it}^O}{\Delta E_{it}} = -\frac{1}{\hat{\beta} + \hat{\theta}} \frac{E_{it}^O}{E_{it}}. \quad (18)$$

That is, the leakage rate is a function of the estimated elasticities and domestic and imported emissions for the firm-years in the sample.

B.2.3 Chinese import competition

We calculate the leakage rate for Chinese imports using our firm-specific Chinese import competition measure, CIP_{it} . The leakage rate is (dropping time subscripts)

$$L_i^{CIP} = -\frac{\Delta E_i^{CN}}{\Delta E_i}. \quad (19)$$

The denominator is calculated straightforwardly as

$$\Delta E_i = e_i^Y \Delta Y_i + Y_i \Delta e_i^Y \quad (20)$$

$$= (\hat{\delta} + \hat{\lambda}) E_i \Delta CIP_i \quad (21)$$

Here we simply use the formula for the estimated coefficients, $\hat{\delta}$ and $\hat{\lambda}$, and plug them into the total differential of $E_i = e_i^Y Y_i$.

The next step is to measure the numerator, ΔE_i^{CN} , the emissions embodied in imports from China that can be allocated to firm i . Note that for each product k , the embodied emissions in imports of final products from China is $E_k = e_k^{CN} M_k^{CN}$, where e_k^{CN} is the Exiobase emission intensity for product k when produced in China and M_k^{CN} is final goods imports from China of product k corresponding to the import flows that enter the numerator of our firm-specific import competition measure in equation (2).

The embodied emissions in total imports from China is $E^{CN} = \sum_k e_k^{CN} M_k^{CN}$, and the change in that variable is then $\Delta E^{CN} = \sum_k e_k^{CN} \Delta M_k^{CN}$. A simple approach to allocate that total change to each Danish firm is to let each firm get a share of each product corresponding to their market shares among Danish firms in a given product over the entire sample window, 1996-2016. That is

$$\Delta E_i^{CN} = \sum_k \sigma_{ik} e_k^{CN} \Delta M_k^{CN}, \quad (22)$$

where σ_{ik} are market shares such that $\sum_i \sigma_{ik} = 1$. This approach means we avoid "double counting" as $\Delta E^{CN} = \sum_i \Delta E_i^{CN}$.

C Additional Tables

Table C1: Firm-level Effects of Offshoring and Chinese Import Competition - OLS

	ln CO ₂ Intensity			ln Sales			ln CO ₂		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ln Off	-0.135*** (0.011)	-0.135*** (0.011)		0.227*** (0.007)	0.227*** (0.007)		0.092*** (0.011)	0.092*** (0.011)	
CIP	-0.008 (0.141)		0.008 (0.142)	-0.238*** (0.076)		-0.265*** (0.085)	-0.246* (0.143)		-0.257* (0.145)
Obs.	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591	11,591

Notes: All columns show OLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C2: Labor Productivity Effects

	ln Value Added per Worker		
	(1)	(2)	(3)
ln Off	0.298** (0.138)	0.299** (0.132)	
CIP	-0.028 (0.308)		-0.325 (0.242)
Observations	11,543	11,543	11,543

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C3: Robustness: Electricity Switching

	ln Electricity (GJ) (1)	ln Non Electricity (GJ) (2)	ln Total (GJ) (3)	ln Electricity Share (4)
ln Off	0.568*** (0.204)	-0.080 (0.247)	0.338* (0.173)	0.230* (0.131)
CIP	-2.154*** (0.461)	-3.533*** (0.619)	-2.821*** (0.400)	0.667** (0.298)
Observations	11,407	11,407	11,407	11,407

Notes: All columns show 2SLS regressions and include year and firm fixed effects. The sample excludes firms in the iron and steel, aluminum, and cement industries. Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C4: Firm-level Effects of Offshoring and Chinese Import Competition with Controls

	ln CO ₂ Intensity (1)	ln Sales (2)	ln CO ₂ Intensity (3)	ln Sales (4)	ln CO ₂ Intensity (5)	ln Sales (6)
ln Off	-0.680** (0.289)	0.620*** (0.172)	-0.802** (0.346)	0.524*** (0.169)	-0.789** (0.381)	0.448** (0.174)
CIP	-1.997*** (0.712)	-1.412*** (0.366)	-1.885** (0.750)	-1.380*** (0.333)	-1.307* (0.723)	-0.513* (0.272)
Exporter dummy	0.038 (0.046)	-0.018 (0.028)				
ln Exports			0.045 (0.032)	0.006 (0.015)		
ln Employment					0.410 (0.292)	0.456*** (0.133)
ln Capital					0.016 (0.034)	0.033* (0.019)
Observations	11,591	11,591	10,463	10,463	11,522	11,522

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C5: CO₂ without process emissions

	ln CO ₂ Int.	ln Sales	ln CO ₂
	(1)	(2)	(3)
ln Off	-0.630** (0.293)	0.617*** (0.168)	-0.013 (0.250)
CIP	-1.752** (0.757)	-1.409*** (0.366)	-3.161*** (0.640)
Observations	11,591	11,591	11,591

Notes: All columns show 2SLS regressions and include year and firm fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C6: Counterfactual decompositions with alternative parameter estimates

Scenario	Total	Technique	Composition	Cross
Panel A: Actual Decomposition				
Actual	-0.55	-0.28	-0.29	0.02
Panel B: Single endogenous regressor				
Counterfactual emission intensity with offshoring at initial level	-0.28	0.00	-0.29	0.01
Counterfactual sales with offshoring at initial level	-0.42	-0.28	-0.13	-0.01
Counterfactual emission intensity with CIP at initial level	-0.55	-0.28	-0.29	0.02
Counterfactual sales with CIP at initial level	-0.57	-0.28	-0.32	0.03
Panel C: Heterogeneity				
Counterfactual emission intensity with offshoring at initial level	-0.18	0.10	-0.29	0.01
Counterfactual sales with offshoring at initial level	-0.65	-0.28	-0.66	0.28
Counterfactual emission intensity with CIP at initial level	-0.55	-0.29	-0.29	0.02
Counterfactual sales with CIP at initial level	-0.56	-0.28	-0.30	0.02

Notes: The table shows decompositions of the relative change in the aggregate emission intensity as defined in equation (9). Panel A replicates the actual decomposition of the balanced panel reported in Table 2. Panel B performs decompositions where values are calculated with parameter estimates from the regressions with one endogenous regressors (Table 3). In Panel C the decompositions are performed with counterfactual emission intensities and sales calculated by using the parameter estimates that allows for heterogeneous effects (Table 6). All decompositions rely on a balanced sample of firms operating in 1996 and 2016, so the net entry effect is zero by construction and not shown. The counterfactual scenarios rely on the method outlined in appendix B.1.

Table C7: Counterfactual decompositions over seven year periods

Scenario	Total	Within	Between	Cross
Panel A: 1996 - 2003				
Actual	-0.23	-0.07	-0.12	-0.04
Counterfactual emission intensity with offshoring at initial level	0.01	0.14	-0.12	-0.01
Counterfactual sales with offshoring at initial level	0.01	-0.07	0.17	-0.09
Counterfactual emission intensity with CIP at initial level	-0.22	-0.06	-0.12	-0.04
Counterfactual sales with CIP at initial level	-0.24	-0.07	-0.13	-0.04
Panel B: 2003 - 2009				
Actual	-0.32	-0.18	-0.13	0.00
Counterfactual emission intensity with offshoring at initial level	-0.14	-0.07	-0.13	0.07
Counterfactual sales with offshoring at initial level	-0.15	-0.18	0.02	0.02
Counterfactual emission intensity with CIP at initial level	-0.31	-0.17	-0.13	-0.01
Counterfactual sales with CIP at initial level	-0.33	-0.18	-0.14	0.00
Panel C: 2009 - 2016				
Actual	-0.27	-0.06	-0.14	-0.07
Counterfactual emission intensity with offshoring at initial level	0.40	0.59	-0.14	-0.05
Counterfactual sales with offshoring at initial level	-0.25	-0.06	-0.16	-0.04
Counterfactual emission intensity with CIP at initial level	-0.27	-0.06	-0.14	-0.07
Counterfactual sales with CIP at initial level	-0.28	-0.06	-0.15	-0.07

Notes: The table shows decompositions of the relative change in the aggregate emission intensity as defined in equation (9). Each panel uses a balanced sample of firms that operate in the indicated period, so the net entry effect is zero by construction. The counterfactual scenarios rely on the methodology explained in appendix B.1. Counterfactual values are calculated with parameter estimates from the regressions with two endogenous variables in Table 3.

Table C8: Carbon Leakage Rates

	Average	Average (weighted)	25th percentile	Median	75th percentile
Offshoring	7.40 (30.26)	6.84 (27.96)	1.64 (6.69)	4.07 (16.65)	8.86 (36.23)
Chinese import penetration	469.62 (81.34)	370.18 (64.12)	12.15 (2.10)	144.87 (25.09)	497.91 (86.24)

Notes: The table calculates leakage rates based on the regression parameter estimates from Table 3, column (7), domestic emissions and imported emissions calculated using Exiobase emission coefficients. The unit of observation is a firm-year. This implies that the underlying calculation has one leakage rate for each firm year. The column denoted 'Average (weighted)' weights each leakage rate by absolute emissions in the firm. Some firms only produce products that cannot be matched to Exiobase products. These are not included in the calculations. The top and bottom percentiles of leakage rates have been trimmed. Standard errors in parentheses are calculated using the delta method.

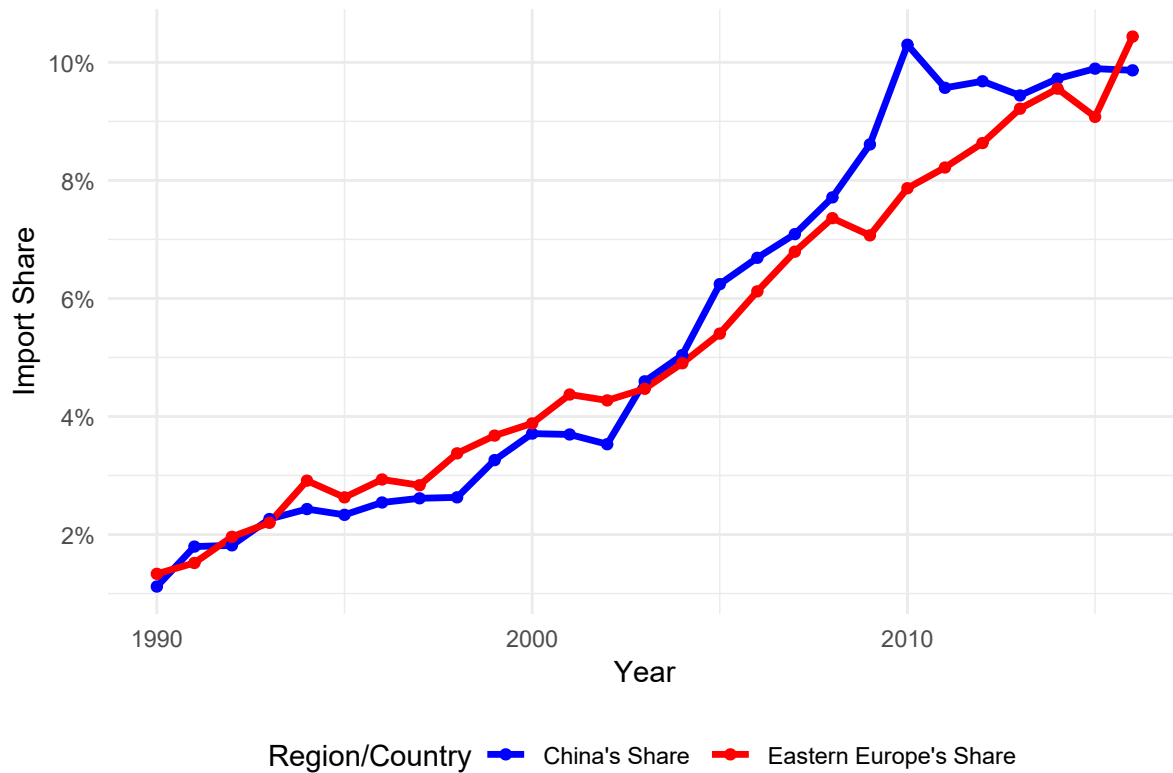
Table C9: Leakage Rates for Chinese Import Competition by industry

2-digit NACE industry	Avg.	Weighted avg. (CO2)	Median	N
Food Products	464.04	461.25	24.29	713
Beverages	78.88	69.80	17.66	43
Tobacco				10
Textiles	380.89	546.69	67.87	196
Wearing Apparel	1094.09	1059.62	197.92	106
Leather	1648.88	717.88	313.63	24
Wood	1165.43	500.57	197.52	272
Paper	247.10	494.41	95.12	205
Media Printing				11
Coke and Refined Petroleum				4
Chemicals	937.15	1652.55	160.07	297
Pharmaceuticals	1636.45	2499.41	61.18	95
Rubber and Plastic	585.31	1003.68	234.91	760
Non-Metallic Mineral	890.61	317.44	91.25	305
Basic Metals	919.13	550.21	382.19	130
Metal Products	634.16	231.10	226.64	967
Electronic Products	339.94	210.82	131.94	492
Electrical Equipment	792.39	986.67	250.28	266
Machinery	418.15	44.32	113.86	1615
Motor Vehicles	719.33	509.97	222.83	211
Other Transportation	1255.51	1245.33	401.38	56
Furniture	470.15	367.33	177.02	406
Other	931.41	1498.13	290.97	233

Notes: The table shows leakage rates for Chinese import competition by 2-digit NACE industries. The leakage rates are calculated using emission coefficients from Exiobase.

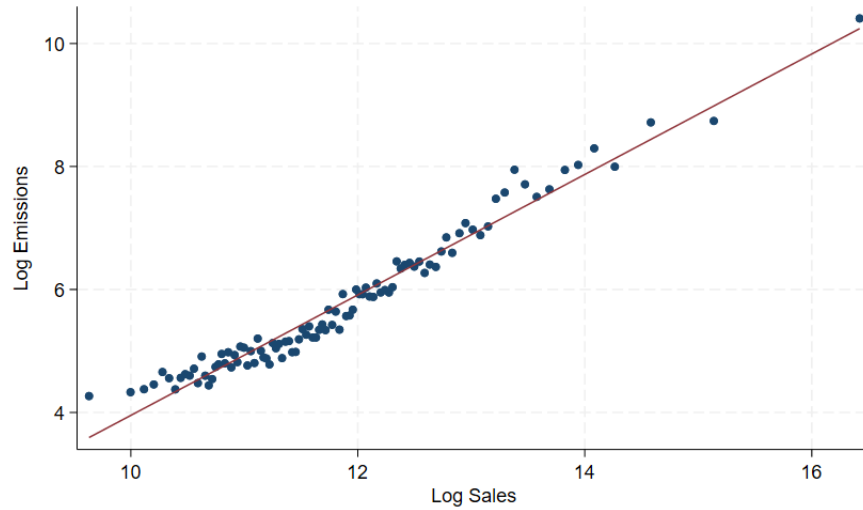
D Additional Figures

Figure D1: Country Shares in Denmark's Manufacturing Imports



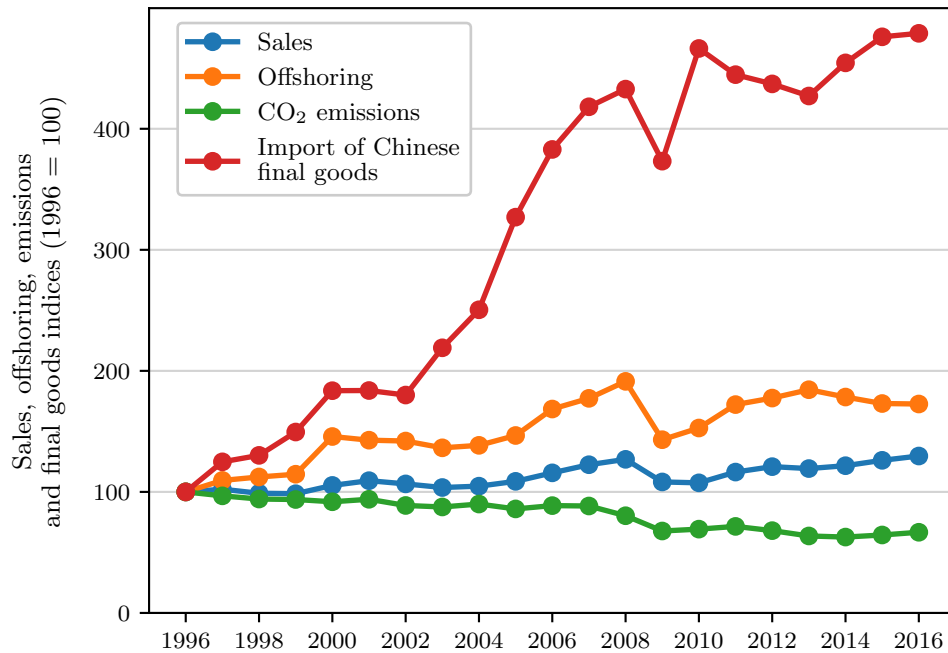
Notes: The figure shows shares of total Danish manufacturing goods imports for China and the Central and Eastern European countries (CEEC) that became members of the EU in 2004 and 2007. The shares are computed as imports of manufacturing goods relative to total imports of manufacturing goods from the World. The source data is WITS using the SITC Revision 1 classification to identify manufacturing products.

Figure D2: Scatterplot of emission against sales



Notes: The figures shows a binned scatter plot. The regression line is based on all underlying observations ($N = 11591$) and controls for year fixed effects. The regression line has a slope of 0.98 ($t = 86.95$).

Figure D3: Sales, offshoring and emissions in Danish manufacturing and imports of Chinese final goods



Notes: The figure shows aggregate emissions reported from Statistics Denmark as well as offshoring, production and imports of final goods from China calculated from firm-level data. The micro data consists of observations on all Danish manufacturing firms with 20 or more employees. All monetary variables are deflated to year 2015. Offshoring measures the imports of products that firms could likely have produced themselves. Final goods are imports by non-manufacturing firms. The emission index is calculated based on the aggregate domestic emissions from manufacturing including biomass.